

To increase our knowledge the subject [oceanography] must be made attractive to men who do not mind facing up to the difficulties of fluid mechanics.

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CHAPTER FOURTEEN

Wind-Driven Gyres

UNDERSTANDING THE CIRCULATION OF THE OCEAN involves a combination of observations, comprehensive numerical modelling, and more conceptual modelling, or ‘theory’. All have become essential, but in this chapter, and the ones following, our emphasis is on the last of the tripos. Its (continuing) role is not to explain every feature of the observed ocean circulation, nor to necessarily describe details best left to numerical simulations. It is to provide a conceptual and theoretical framework for understanding the circulation of the ocean, for interpreting observations and suggesting how new observations may best be made, and to aid the development and interpretation of experiments with numerical models.

The main currents of the world's oceans are sketched in Fig. 14.1.¹ (Over most of the ocean, the vertically averaged currents have a similar sense as the surface currents, one exception being at the equator where the surface currents are mainly westwards but the vertical integral is dominated by the eastwards undercurrent.) Two seemingly dichotomous aspects stand out: (i) the complexity of the currents as they interact with topography and the geography of the continents; (ii) the simplicity and commonality of the large-scale structures in the major ocean basins, and in particular the ubiquity of subtropical and subpolar gyres. Indeed these gyres, sweeping across the great oceans carrying vast quantities of water and heat, are perhaps the single most conspicuous feature of the circulation. The subtropical gyres are anticyclonic, extending polewards to about 45°, and the subpolar gyres are cyclonic and polewards of this, primarily in the Northern Hemisphere. The existence of gyres, and that they are strongest in the west, has been known for centuries; this *western intensification* leads to such well-known currents as the Gulf Stream in the Atlantic (charted by Benjamin Franklin), the Kuroshio in the Pacific, and the Brazil Current in the South Atlantic. Although even today we barely have sufficient observations to produce a

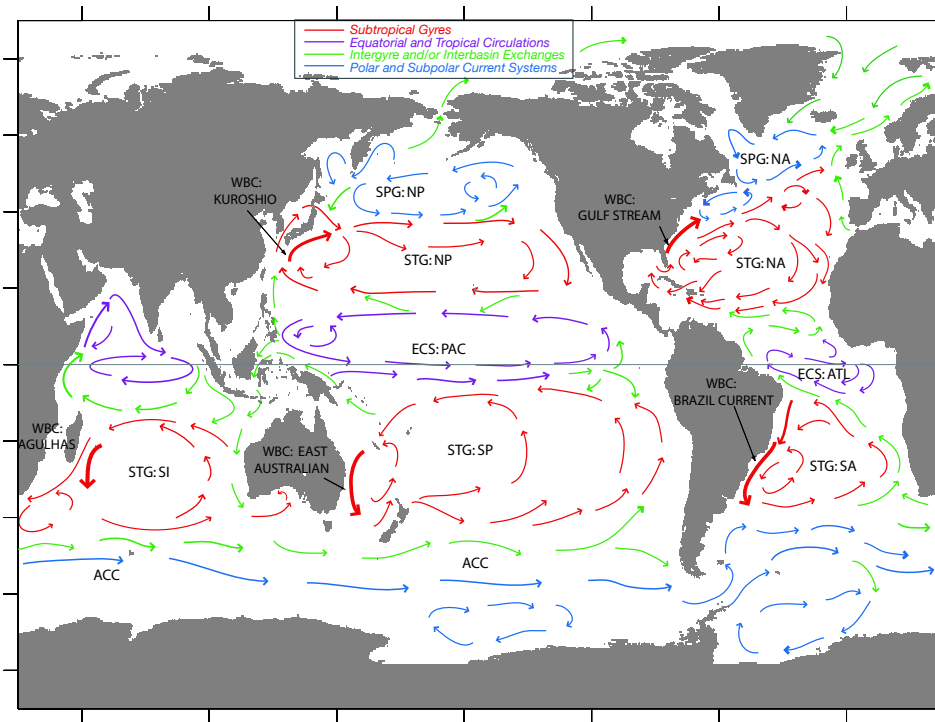


Fig. 14.1 A schema of the main currents of the global ocean. Key: STG – Sub-Tropical Gyre; SPG – Sub-Polar Gyre; WBC – Western Boundary Current; ECS – Equatorial Current System; NA – North Atlantic; SA – South Atlantic; NP – North Pacific; SP – South Pacific; SI – South Indian; ACC – Antarctic Circumpolar Current; ATL – Atlantic; PAC – Pacific. The figure is a qualitative, and not quantitative, representation of the actual flow.

synoptic map of the ocean currents, except at the surface, the large-scale mean currents are fairly well mapped and Fig. 14.2 illustrates the average current pattern of the North Atlantic using a combination of observations and a numerical model, and the Gulf Stream is clearly visible.²

For much of this chapter we consider a model, and variations about it, that explains the large-scale features of ocean gyres and that lies at the core of ocean circulation theory — the steady, forced-dissipative, homogeneous model of the ocean circulation first formulated by Stommel.⁴ Such models explain many of the zeroth-order features of the ocean, in particular the existence of gyres and the appearance of intense western boundary currents. In the later part of the chapter we examine the vertical structure of the wind-driven circulation, taking the stratification as given. In the two chapters following we consider the maintenance of that stratification.

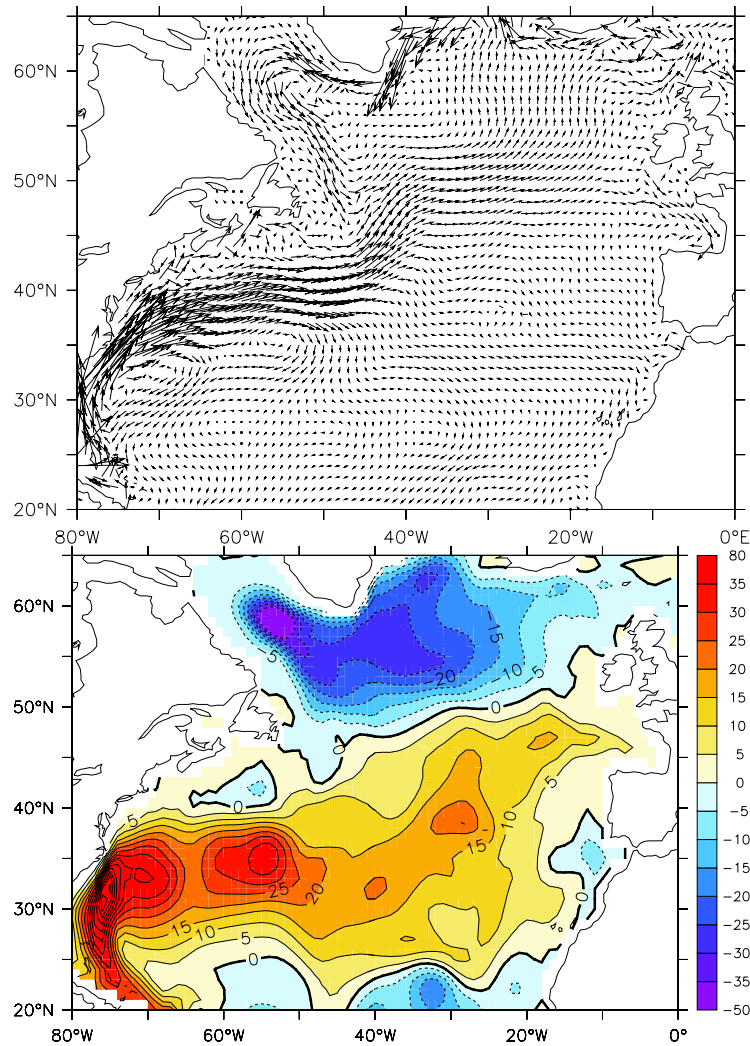


Fig. 14.2 Top: The time-averaged velocity field at a depth of 75 m in the North Atlantic, obtained by constraining a numerical model to observations. Bottom: The streamfunction of the vertically integrated flow, in Sverdrups ($1 \text{ Sv} = 10^9 \text{ kg s}^{-1}$). Note the presence of an anticyclonic subtropical gyre, a cyclonic subpolar gyre, and intense western boundary currents.³

14.1 THE DEPTH INTEGRATED WIND-DRIVEN CIRCULATION

The equations that govern the large-scale flow in the oceans are the planetary-geostrophic equations. Greatly simplified as these are compared to the Navier-Stokes equations, or even the hydrostatic Boussinesq equations, they are still quite daunting: a prognostic equation for buoyancy is coupled to the advecting velocity via hydrostatic and geostrophic balance, and the resulting problem is formidably nonlinear. However, it turns out that thermodynamic effects can effectively be eliminated by the simple device of vertical integration; the resulting equations are linear, and the only external

forcing is that due to the wind-stress. The resulting model then, at the price of some comprehensiveness, gives a useful picture of the *wind-driven* circulation of the ocean.

14.1.1 The Stommel Model

The planetary geostrophic equations for a Boussinesq fluid are:

$$\frac{Db}{Dt} = \dot{b}, \quad \nabla_3 \cdot \mathbf{v} = 0, \quad (14.1a,b)$$

$$\mathbf{f} \times \mathbf{u} = -\nabla \phi + \frac{1}{\rho_0} \frac{\partial \boldsymbol{\tau}}{\partial z}, \quad \frac{\partial \phi}{\partial z} = b. \quad (14.2a,b)$$

These equations are, respectively, the thermodynamic equation (14.1a), the mass continuity equation (14.1b), the horizontal momentum equation (14.2a), (i.e., geostrophic balance, plus a stress term) and the vertical momentum equation (14.2b), (i.e., hydrostatic balance). These equations are derived more fully in chapter 5, but they are essentially the Boussinesq primitive equations with the advection terms omitted in the horizontal momentum equation, on the basis of small Rossby number. In this chapter we will absorb the factor of ρ_0 into $\boldsymbol{\tau}$, so that $\boldsymbol{\tau}$ is kinematic stress, stress/ ρ_0 , and the gradient operator will be two dimensional, in the x - y plane, unless noted.

Take the curl of (14.2a) (that is, cross differentiate its x and y components) and integrate over the depth of the ocean to give

$$\int \mathbf{f} \nabla \cdot \mathbf{u} \, dz + \frac{\partial f}{\partial y} \int v \, dz = \text{curl}_z(\boldsymbol{\tau}_T - \boldsymbol{\tau}_B), \quad (14.3)$$

where the operator curl_z is defined by $\text{curl}_z \mathbf{A} \equiv \partial A^y / \partial x - \partial A^x / \partial y = \mathbf{k} \cdot \nabla \times \mathbf{A}$, and the subscripts t and b are for top and bottom. The divergence term vanishes if the vertical velocity is zero at the top and bottom of the ocean. Strictly, at the top of the ocean the vertical velocity is given by the material derivative of height of the ocean's surface, Dh/Dt , but on the large-scales this has a negligible effect and we may make the rigid-lid approximation and set it to zero. At the bottom of the ocean the vertical velocity is only zero if the ocean is flat-bottomed; otherwise it is $\mathbf{u} \cdot \nabla \eta_B$, where η_B is the orographic height at the ocean floor. The neglect of this topographic term is probably the most restrictive single approximation in the model. Given this neglect, (14.3) becomes

$$\beta \bar{v} = \text{curl}_z(\boldsymbol{\tau}_T - \boldsymbol{\tau}_B) \quad (14.4)$$

where henceforth, in this section, quantities with an overbar are understood to be the vertical integral over the depth of the ocean. If the stresses depend only on the velocity fields then thermodynamic fields do not affect the vertically integrated flow.

At the top of the ocean, the stress is given by the wind. At the bottom, in the absence of topography we assume that the stress may be parameterized by a linear drag, or Rayleigh friction, as might be generated by an Ekman layer; it is this assumption that particularly characterizes this model as that due to Stommel. (Note that we parameterize the friction by a drag acting on the vertically integrated velocity. Using the velocity at the bottom of the ocean would be more realistic, but this wrinkle is beyond the scope of vertically integrated models.) Eq. (14.4) then becomes

$$\beta \bar{v} = -r \bar{\zeta} + W_E(x, y) \quad (14.5)$$

Formulation of the Stommel/Munk model

- ★ Vertically integrated planetary-geostrophic equations, or a homogeneous fluid with nonlinearity neglected.
- ★ Friction parameterized by a linear drag (Stommel model) or a harmonic Newtonian viscosity (Munk model) or both (Stommel-Munk model).
- ★ Flat bottomed ocean.

Variations on the theme include allowing nonlinearity in the vorticity equation, posing the problem in domains of various shapes, and allowing bottom topography (in particular sloping sidewalls). The solutions are normally calculated in the boundary-layer approximation, but some exact solutions exist.

where W_E is proportional to the wind-stress curl at the top of the ocean and is a known function. Because the velocity is divergence-free, we can define a streamfunction ψ such that $\bar{u} = -\partial\psi/\partial y$ and $\bar{v} = \partial\psi/\partial x$. Eq. (14.5) then becomes

$$r\nabla^2\psi + \beta\frac{\partial\psi}{\partial x} = W_E(x, y). \quad (14.6)$$

This equation is often referred to as the *Stommel problem* or *Stommel model*, and may be posed in a variety of two dimensional domains.

14.1.2 Alternative formulations

A number of alternative formulations leading to (14.6) are possible. None are perhaps as well justified as the derivation via the planetary-geostrophic equations but the differences in the specific assumptions made both give some indication of the robustness of the derivation, and show how the model might be extended to include topographic or nonlinear effects.

1. A Homogeneous Model

Rather than vertically integrating, we may suppose that the ocean is a homogeneous fluid obeying the shallow water equations (chapter 3). The potential vorticity equation [c.f., (3.80) on page 138] is:

$$\frac{D}{Dt} \left(\frac{\zeta + f}{h} \right) = \frac{F}{h}, \quad (14.7)$$

where F represents friction and forcing [c.f., (3.80)]. In an ocean with a rigid-lid and flat bottom then we obtain the barotropic vorticity equation,

$$\frac{D\zeta}{Dt} + \beta v = F. \quad (14.8)$$

The forcing term F we again represent as a wind-stress curl and a linear drag. Further, since the horizontal velocity is divergence-free (because of the flat-bottom and rigid-lid assumptions) we may represent it as a streamfunction, whence we obtain the closed

equation

$$\frac{D}{Dt} \nabla^2 \psi + \beta \frac{\partial \psi}{\partial x} = W_E(x, y) - r \nabla^2 \psi. \quad (14.9)$$

where W_E represents the wind forcing. This equation is the ‘time-dependent nonlinear Stommel problem’. The steady nonlinear problem is sometimes of interest, too, and this is just

$$J(\psi, \nabla^2 \psi) + \beta \frac{\partial \psi}{\partial x} = W_E(x, y) - r \nabla^2 \psi. \quad (14.10)$$

To obtain the original Stommel model we just ignore the advective derivative, which will be valid if $|\zeta| \sim Z \ll \beta L$ where $Z = U/L$ is a representative value of vorticity. This condition is equivalent to

$$R_\beta \equiv \frac{U}{\beta L^2} \ll 1. \quad (14.11)$$

R_β is called the *beta-Rossby number*. On sufficiently large scales, $\beta \sim f/L$ and (14.11) is similar to a small Rossby number assumption.

II. Quasi-geostrophic formulation

In the PG formulation, the horizontal velocity is divergence-free only because we have vertically integrated. If the scales of motion are not too large the horizontal flow at every level is divergence-free for another reason: because it is in geostrophic balance. In reality, over a single oceanic gyre (say from 15° to 40° latitude), variations in Coriolis parameter are not large, and this prompts us to formulate the model in terms of the quasi-geostrophic equations. Formally, such a model would then be restricted to length-scales L no more than $\mathcal{O}(Ro^{-1})$ larger than the deformation radius, and for gyre scales this criterion is marginally satisfied if $L_d = 100$ km. An advantage of the quasi-geostrophic equations is that it readily allows for the inclusion of both nonlinearity and stratification. [For an informal summary of quasi-geostrophy, see the box on page 224.] The quasi-geostrophic potential vorticity equation for a Boussinesq system is:

$$\frac{D}{Dt} \left[\zeta + \frac{\partial}{\partial z} \left(\frac{f_0^2}{N^2} \frac{\partial \psi}{\partial z} \right) \right] + \beta v = f_0 \frac{\partial w}{\partial z} + \text{curl}_z \frac{\partial \tau}{\partial z} \quad (14.12)$$

If we neglect the advective derivative, and vertically integrate, we obtain

$$\beta \int v \, dz = f_0 [w]_B^T + \text{curl}_z [\tau]_B^T. \quad (14.13)$$

where T denotes the ocean top and B the bottom. We now make one of two virtually equivalent choices.

- (i) We suppose that the integration is over the entire depth of the ocean, in which case the term $[w]_B^T$ vanishes given a rigid lid and a flat bottom. If the stress at the top of the ocean is given by the wind stress, and at the bottom of the ocean it is parameterized by a linear drag, then we obtain

$$\beta \bar{v} = W_E(x, y) - r \bar{\zeta} \quad (14.14)$$

just as in (14.5), and where an overbar denotes a vertical integral. Writing $\bar{v} = \partial \psi / \partial x$ and $\bar{\zeta} = \nabla^2 \psi$ then gives (14.6).

- (ii) We suppose that the integration is between two thin Ekman layers at the top and bottom of the ocean. The stress is zero at the interior edge of these layers, but the vertical velocity is not. At the base of the upper Ekman layer it is given by:

$$w(x, y, -\delta_u) = \text{curl}_z(\boldsymbol{\tau}_T / f_0) \quad (14.15)$$

where the top of the ocean is at $z = 0$ and δ_u is the thickness of the upper Ekman layer. Similarly, at the top of the lower Ekman layer, the vertical velocity is:

$$w(x, y, -H + \delta_B) = \delta_B \zeta \quad (14.16)$$

where $z = -H$ at the ocean bottom and δ_B is the thickness of the bottom Ekman layer. Neglecting the advective derivative, and integrating over the ocean between the two Ekman layers, (14.12) becomes

$$\beta \bar{v} = \text{curl}_z \boldsymbol{\tau}_T - f_0 \delta_B \bar{\zeta}. \quad (14.17)$$

Defining the drag coefficient r by $r = f_0 \delta_B$, and introducing a streamfunction gives, as before,

$$r \nabla^2 \psi + \beta \frac{\partial \psi}{\partial x} = \text{curl}_z \boldsymbol{\tau}_T. \quad (14.18)$$

14.1.3 Approximate solution of Stommel model

Sverdrup balance

Eq. (14.6) is linear and it is possible to obtain an exact, analytic solution. However, it is more insightful to approach the problem perturbatively, by supposing that the frictional term is small, meaning there is an approximate balance between wind-stress and the β -effect.⁵ Friction is small if $|r\zeta| \ll |\beta v|$ or

$$\frac{r}{L} = \frac{f\delta}{HL} \ll \beta \quad (14.19)$$

using $r = fd/H$ where d is the thickness of the Ekman layer and L is the horizontal scale of the motion, and generally speaking this inequality is well satisfied for large-scale flow. The vorticity equation becomes

$$\beta \bar{v} \approx \text{curl}_z \boldsymbol{\tau}_T, \quad (14.20)$$

or *Sverdrup balance*.⁶ The observational support for Sverdrup balance is rather mixed, discrepancies arising not so much from failure of (14.19), but from the presence of small-scale eddying motion with concomitantly large nonlinear terms, and the presence of non-negligible vertical velocities induced by the interaction with bottom topography.⁷ Nevertheless, Sverdrup balance provides a useful, if not impregnable, foundation on which to build. (Sometimes ‘Sverdrup balance’ is taken to mean the linear geostrophic vorticity balance $\beta v = f \partial w / \partial z$, but we will restrict its use to a balance between the beta effect and wind stress curl.)

Boundary-layer solution

For simplicity, consider a square domain of side a and rescale the variables by setting

$$x = a\hat{x}, \quad y = a\hat{y}, \quad \tau = \tau_0\hat{\tau}, \quad \psi = \hat{\psi}\frac{\tau_0}{\beta} \quad (14.21)$$

where τ_0 is the amplitude of the windstress. The hatted variables are nondimensional and, assuming our scaling to be sensible, these are $\mathcal{O}(1)$ quantities in the interior. Eq. (14.18) becomes

$$\frac{\partial \hat{\psi}}{\partial \hat{x}} + \epsilon_S \nabla^2 \hat{\psi} = \text{curl}_z \hat{\tau}_T, \quad (14.22)$$

where $\epsilon_S = (r/a\beta) \ll 1$, in accord with (14.19). For the rest of this section we will drop the hats over nondimensional quantities. Over the interior of the domain, away from boundaries, the frictional term in (14.22) is small. We can take advantage of this by writing

$$\psi(x, y) = \psi_I(x, y) + \phi(x, y) \quad (14.23)$$

where ψ_I is the interior streamfunction and ϕ is a boundary layer correction. Away from boundaries ψ_I is presumed to dominate the flow, and this satisfies

$$\frac{\partial \psi_I}{\partial x} = \text{curl}_z \tau_T. \quad (14.24)$$

The solution of this equation (called the ‘Sverdrup interior’) is

$$\psi_I(x, y) = \int_0^x \text{curl}_z \tau(x', y) dx' + g(y), \quad (14.25)$$

where $g(y)$ is an arbitrary function of y , which thus gives rise to an arbitrary zonal flow. The corresponding velocities are

$$v_I = \text{curl}_z \tau, \quad u_I = -\frac{\partial}{\partial y} \int_0^x \text{curl}_z \tau(x', y) dx' - \frac{\partial g(y)}{\partial y}. \quad (14.26)$$

The dynamics is most clearly illustrated if we now restrict attention to a windstress curl that is zonally uniform, and that vanishes at two latitudes, $y = 0$ and $y = 1$. An example is

$$\tau_T^y = 0, \quad \tau_T^x = -\tau_0 \cos(\pi y), \quad (14.27)$$

for which $\text{curl}_z \tau_T = -\pi \sin(\pi y)$. The Sverdrup (interior) flow is then

$$\psi_I(x, y) = (x - C(y)) \text{curl}_z \tau_T = \pi(C(y) - x) \sin \pi y. \quad (14.28)$$

If we choose C to be a constant, the arbitrary zonal flow is $C \text{curl}_z \tau_T$. We can then satisfy $\psi = 0$ at *either* $x = 0$ (if $C = 0$) *or* $x = 1$ (if $C = 1$). These solutions are illustrated in Fig. 14.3 for the particular stress (14.27).

Regardless of our choice of C we cannot satisfy $\psi = 0$ at both zonal boundaries. We must choose one, and then construct a *boundary layer* solution (i.e., determine ϕ) to satisfy the other condition. Which choice do we make? On intuitive grounds it seems that we should choose the solution that satisfies $\psi = 0$ at $x = 1$ (the solution on the left in Fig. 14.3), for then the interior flow then goes round in the same

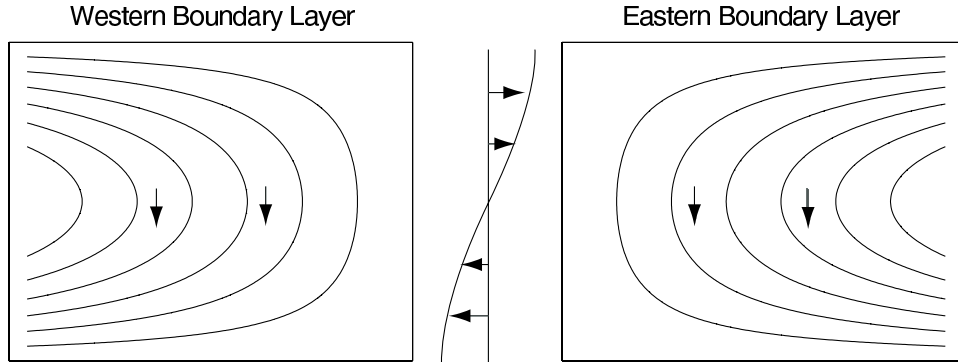


Fig. 14.3 Two possible Sverdrup flows, ψ_I , for the wind stress shown in the center. Each solution satisfies the no-flow condition at either the eastern or western boundary, and a boundary layer is therefore required at the other boundary. Both flows have the same, equatorward, meridional flow in the interior. Only the flow with the western boundary current is physically realizable, however, because only then can friction produce a torque that opposes that of the wind-stress, so allowing the flow to equilibrate.

direction as the wind: the wind is supplying a clockwise torque, and to achieve an angular momentum balance anticlockwise angular momentum must be removed by friction. We can imagine that this would be provided by the frictional forces at the western boundary layer if the interior flow is clockwise, but not by friction at an eastern boundary layer when the interior flow is anticlockwise. Note that this argument is not dependent on the sign of the windstress-curl: if the wind blew the other way a similar argument still implies that a western boundary layer is needed. We will now see if and how the mathematics reflects this intuitive but non-rigorous argument.

Asymptotic Matching

Near the walls of the domain the boundary layer correction $\phi(x, y)$ must become important in order that the boundary conditions may be satisfied, and the flow, and in particular $\phi(x, y)$, will vary rapidly with x . To reflect this, let us *stretch* the x -coordinate near this point of failure (i.e., at $x = 0$ or $x = 1$, but we don't know which yet) and let

$$x = \epsilon\alpha \quad \text{or} \quad x - 1 = \epsilon\alpha \quad (14.29a,b)$$

Here, α is the stretched-coordinate, which has values $\mathcal{O}(1)$ in the boundary layer, and ϵ is a small parameter, as yet undetermined. We then suppose that $\phi = \phi(\alpha, y)$, and using (14.23) in (14.22), we obtain

$$\epsilon_S (\nabla^2 \psi_I + \nabla^2 \phi) + \frac{\partial \psi_I}{\partial x} + \frac{1}{\epsilon} \frac{\partial \phi}{\partial \alpha} = \text{curl}_z \tau_T, \quad (14.30)$$

where $\phi = \phi(\alpha, y)$ and $\nabla^2 \phi = \epsilon^{-2} \partial^2 \phi / \partial \alpha^2 + \partial^2 \phi / \partial y^2$. Now, by choice, ψ_I exactly satisfies Sverdrup balance, and so (14.30) becomes

$$\epsilon_S \left(\nabla^2 \psi_I + \frac{1}{\epsilon^2} \frac{\partial^2 \phi}{\partial \alpha^2} + \frac{\partial^2 \phi}{\partial y^2} \right) + \frac{1}{\epsilon} \frac{\partial \phi}{\partial \alpha} = 0. \quad (14.31)$$

We now choose ϵ to obtain a physically meaningful solution. The obvious choice is $\epsilon = \epsilon_S$, for then the leading-order balance in (14.31) is

$$\frac{\partial^2 \phi}{\partial \alpha^2} + \frac{\partial \phi}{\partial \alpha} = 0, \quad (14.32)$$

the solution of which

$$\phi = A(\gamma) + B(\gamma) e^{-\alpha}. \quad (14.33)$$

Evidently, ϕ grows exponentially in the negative α direction. If this were allowed, it would violate our assumption that solutions are small in the interior, and we must eliminate this possibility by allowing α to take only positive values in the interior of the domain, and by setting $A(\gamma) = 0$. We therefore choose $x = \epsilon \alpha$ so that $\alpha > 0$ for $x > 0$; the boundary layer is then at $x = 0$, that is, it is a *western boundary*, and it decays eastward in the direction of increasing α — that is, into the ocean interior. We now choose $C = 1$ in (14.28) to make $\psi_I = 0$ at $x = 1$ in (14.28) and then, for the windstress (14.27), the interior solution is given by

$$\psi_I = \pi(1 - x) \sin \pi y. \quad (14.34)$$

This alone satisfies the boundary condition at the eastern boundary. The function $B(\gamma)$ is chosen to satisfy the additional condition that

$$\psi = \psi_I + \phi = 0 \quad \text{at } x = 0, \quad (14.35)$$

and using (14.34) this gives

$$\pi \sin \pi y + B(\gamma) = 0. \quad (14.36)$$

Using this in (14.33), with $A(\gamma) = 0$, then gives the boundary layer solution

$$\phi = -\pi \sin \pi y e^{-x/\epsilon_S}. \quad (14.37)$$

The composite (boundary layer plus interior) solution is the sum of (14.34) and (14.37), namely

$$\boxed{\psi = (1 - x - e^{-x/\epsilon_S}) \pi \sin \pi y}. \quad (14.38)$$

With dimensional variables this is

$$\psi = \frac{\tau_0 \pi}{\beta} \left(1 - \frac{x}{a} - e^{-x/(a\epsilon_S)} \right) \sin \frac{\pi y}{a}. \quad (14.39)$$

This is a ‘single gyre’ solution. Two or more gyres can be obtained with a different wind forcing, such as $\tau^x = -\tau_0 \cos(2\pi y)$, as in Fig. 14.4.

It is a relatively straightforward matter to generalize to other wind stresses, provided these also vanish at the two latitudes between which the solution is desired. It is left as a problem to show that in general

$$\psi_I = \int_{x_E}^x \text{curl}_z \boldsymbol{\tau}(x', y) dx', \quad (14.40)$$

and that the composite solution is

$$\psi = \psi_I - \psi_I(0, y) e^{-x/(x_E \epsilon_S)}. \quad (14.41)$$

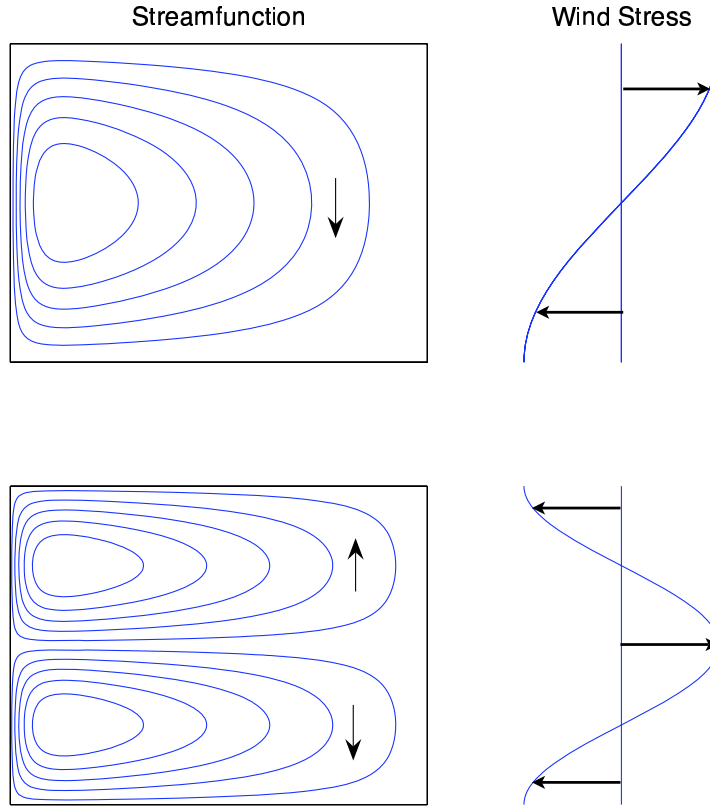


Fig. 14.4 Two solutions of the Stommel model. Upper panel shows the streamfunction of a single-gyre solution, with a windstress proportional to $-\cos(\pi y/a)$ (in a domain of side a), and the lower panel shows a two-gyre solution, with windstress proportional to $\cos(2\pi y/a)$. In both cases $\epsilon_S = 0.04$.

14.2 USING VISCOSITY INSTEAD OF DRAG

A natural variation on the Stommel problem is to use a harmonic viscosity, $\nu \nabla^2 \zeta$, in place of the drag term $-r\zeta$ in the vorticity equation, the argument being that the wind-driven circulation does not reach all the way to the ocean bottom so that an Ekman drag is not appropriate. This variation is called the ‘Munk problem’ or ‘Munk model’,⁸ and if both drag and viscosity are present we have the ‘Stommel-Munk’ model. The particular form of the lateral friction used in the Munk problem is still somewhat hard to justify because it relies on an ill-founded eddy diffusion of relative vorticity (chapter 10). Our treatment of this problem is relatively brief, concentrating only on those areas where the problem differs from the Stommel problem. The problem is to find and understand the solution to the (dimensional) equation

$$\beta \frac{\partial \psi}{\partial x} = \text{curl}_z \tau_T + \nu \nabla^2 \zeta = \text{curl}_z \tau_T + \nu \nabla^4 \psi \quad (14.42)$$

in a given domain, for example a square of side a . We need two boundary conditions at each wall to solve the problem uniquely, and as before for one of them we choose $\psi =$

Some Properties of the Stommel Model

- (i) The transport in the Sverdrup interior is equatorwards for an anti-cyclonic wind-stress curl. This transport is exactly balanced by the polewards transport in the boundary layer.
- (ii) There must be a boundary layer to satisfy mass conservation, and this must be a western boundary layer if the friction acts to provide a torque of opposite sign to the motion itself. As there is a balance between friction and the β -effect, it is a 'frictional boundary layer'. The western location does not depend on the sign of the wind-stress, nor on the sign of the Coriolis parameter, but it does depend on the sign of β , and so on the direction of rotation of the earth.
- (iii) The boundary layer width arises by noting that the terms $r\nabla^2\psi$ and $\beta\partial\psi/\partial x$ are in approximate balance in the western boundary layer, implying a length-scale of $L_W = (r/\beta)$. If r , the inverse frictional time, is 20 days^{-1} then $L_W \approx 60 \text{ km}$, similar to the width of the Gulf Stream.

0 to satisfy the no-normal-flow condition. For the other condition, two possibilities present themselves:

- (i) Zero vorticity, or $\zeta = 0$. Since $\psi = 0$ along the boundary, this possibility is equivalent to $\partial^2\psi/\partial n^2 = 0$ where $\partial/\partial n$ denotes a derivative normal to the boundary. This is known as a 'free-slip' condition. At $x = 0$, for example, the condition becomes $\partial v/\partial x = 0$; that is, there is no horizontal shear at the boundary.
- (ii) No flow along the boundary (the 'no slip' condition). That is $\psi_n = 0$ where the subscript denotes the normal derivative of the streamfunction. At $x = 0$ we have $v = 0$.

There is little a priori justification for choosing either of these. The second choice would be demanded if ν were a molecular viscosity, but then we would have to resolve a molecular boundary layer which perhaps would be a few millimeters thick. Instead, ν must be interpreted as some form of eddy viscosity, as we discuss in chapter 10. In that case, one might argue that the free-slip condition should be preferred, but in the absence of a proper theoretical basis for such eddy viscosities there is no truly rational way to make the choice. We will solve the no-slip problem; see exercise 14.3 for the free-slip problem.

Let the wind stress be the canonical $\tau^x = -\tau_0 \cos(\pi y/a)$. Then the interior (Sverdrup) flow is given by (14.34), as for the Stommel problem. This satisfies the free-slip boundary conditions at $y = 0, 1$, namely $\partial_x^2\psi = 0$, automatically. However, we will need boundary layers at both western and eastern boundaries, because the interior solution cannot satisfy all four boundary conditions required by (14.42). The eastern boundary layer will be relatively weak, and needed only to satisfy the no-slip

or free-slip condition but, as in the Stommel problem, there will be a strong western boundary layer, needed to satisfy the no-normal flow condition. How thick will this be? Inspection of (14.42) suggests that the frictional term and the β term will balance in a boundary layer of thickness of order L_M where

$$L_M = \left(\frac{\nu}{\beta} \right)^{1/3}. \quad (14.43)$$

Nondimensionalizing (14.42) in a similar way as was done for the Stommel problem yields

$$-\epsilon_M \nabla^4 \hat{\psi} + \frac{\partial \hat{\psi}}{\partial \hat{x}} = \text{curl}_z \hat{\tau}_T \quad (14.44)$$

where $\epsilon_M = (\nu/\beta a^3)$. Considering only the western boundary layer correction, we let the solution be the sum of an interior (Sverdrup) streamfunction plus a boundary layer correction:

$$\hat{\psi} = \psi_I + \phi_W(\alpha, \hat{y}) \quad (14.45)$$

where α is a stretched coordinate such that $\hat{x} = \epsilon \alpha$ where ϵ is some small parameter. Substituting (14.45) into (14.44) and subtracting the Sverdrup balance gives

$$-\epsilon_M \left(\nabla^4 \psi_I + \frac{1}{\epsilon^4} \frac{\partial^4 \phi_W}{\partial \alpha^4} \right) + \frac{1}{\epsilon} \frac{\partial \phi_W}{\partial \alpha} = 0. \quad (14.46)$$

A nontrivial balance is obtained when $\epsilon = \epsilon_M^{1/3}$, implying a dimensional western boundary thickness of $\epsilon a = (\nu/\beta)^{1/3}$, consistent with (14.43). Eq. (14.46) then becomes, at leading order,

$$-\frac{\partial^4 \phi_W}{\partial \alpha^4} + \frac{\partial \phi_W}{\partial \alpha} = 0 \quad (14.47)$$

The boundary conditions on (14.47) are that:

- (i) $\phi_W \rightarrow 0$ as $\alpha \rightarrow \infty$. This states that the perturbation decays as it extends into the interior.
- (ii) $\phi_W = -\psi_I$ at $\hat{x} = \alpha = 0$. This is the no-normal-flow condition on the meridional boundary.
- (iii) $\partial \phi_W / \partial \hat{x} = -\partial \psi_I / \partial \hat{x}$ at $\hat{x} = \alpha = 0$. This is the no-slip condition.

In addition, zonal boundary layers must exist at $\hat{y} = 0, 1$ and another meridional boundary layer must exist at $\hat{x} = 1$, in order to satisfy the no slip condition. Solving the full problem is a straightforward albeit non-trivial algebraic exercise and, omitting the weak zonal boundary layers at $\hat{y} = 0, 1$ but including the eastern boundary layer correction, we eventually find

$$\hat{\psi} = \psi_I - \psi_I(0, \hat{y}) e^{-\hat{x}/(2\epsilon)} \left[\cos \left(\frac{\sqrt{3}\hat{x}}{2\epsilon} \right) + \frac{1}{\sqrt{3}} \sin \left(\frac{\sqrt{3}\hat{x}}{2\epsilon} \right) \right] - \epsilon e^{(\hat{x}-1)/\epsilon} \frac{\partial}{\partial \hat{x}} \psi_I(1, \hat{y}). \quad (14.48)$$

where $\epsilon = \epsilon_M^{1/3}$. With the canonical windstress (14.27) this solution becomes

$$\hat{\psi} = \pi \sin(\pi \hat{y}) \left\{ 1 - \hat{x} - e^{-\hat{x}/(2\epsilon)} \left[\cos \left(\frac{\sqrt{3}\hat{x}}{2\epsilon} \right) + \frac{1}{\sqrt{3}} \sin \left(\frac{\sqrt{3}\hat{x}}{2\epsilon} \right) \right] + \epsilon e^{(\hat{x}-1)/\epsilon} \right\} \quad (14.49)$$

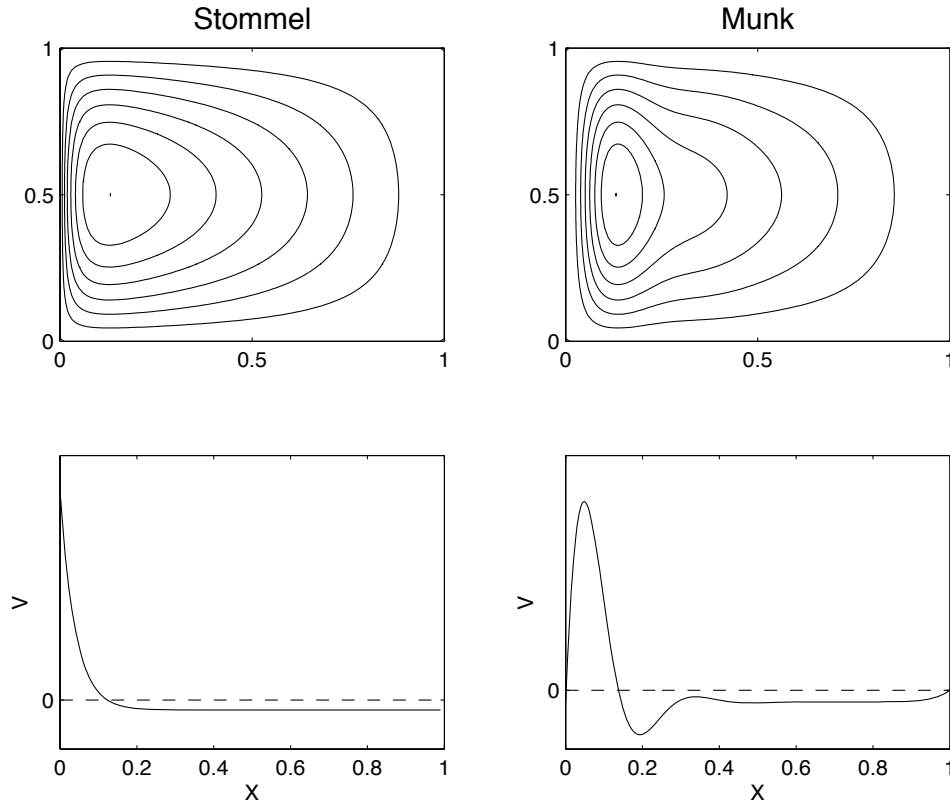


Fig. 14.5 The Stommel and Munk solutions, (14.49) with $\epsilon_S = \epsilon_M^{1/3} = 0.04$, with the windstress $\tau = -\cos \pi y$, for $x, y \in (0, 1)$. Upper panels are contours of streamfunction in the x - y plane, and the flow is clockwise. The lower panels are plots of meridional velocity as a function of x , in the center of the domain ($y = 0.5$). The Munk solution can satisfy both no-normal flow and one other boundary condition at each wall, here chosen to be no-slip.

The solutions of this are plotted in Fig. 14.5. Note how the Munk layers bring the tangential as well as the normal velocity to zero. The eastern boundary layer has similar thickness to the western boundary layer, but is not as dynamically important since its *raison d'être* is to enable the no-slip condition to be satisfied, a relatively weak frictional constraint that manifests itself by a boundary layer in which the flow parallel to the boundary is slowed down. On the other hand the western boundary layer exists in order that the no-normal flow condition can be satisfied, which causes a qualitative change in the flow pattern. It should be emphasized that neither the Stommel nor Munk models are quantitative descriptors of the real ocean, but taken together the similarities of their solutions are a powerful argument for the relative insensitivity of the qualitative form of the solution to the detailed form of the friction. The models do in fact produce qualitatively realistic patterns of large-scale flow in the major basins of the world, as illustrated in Fig. 14.6.

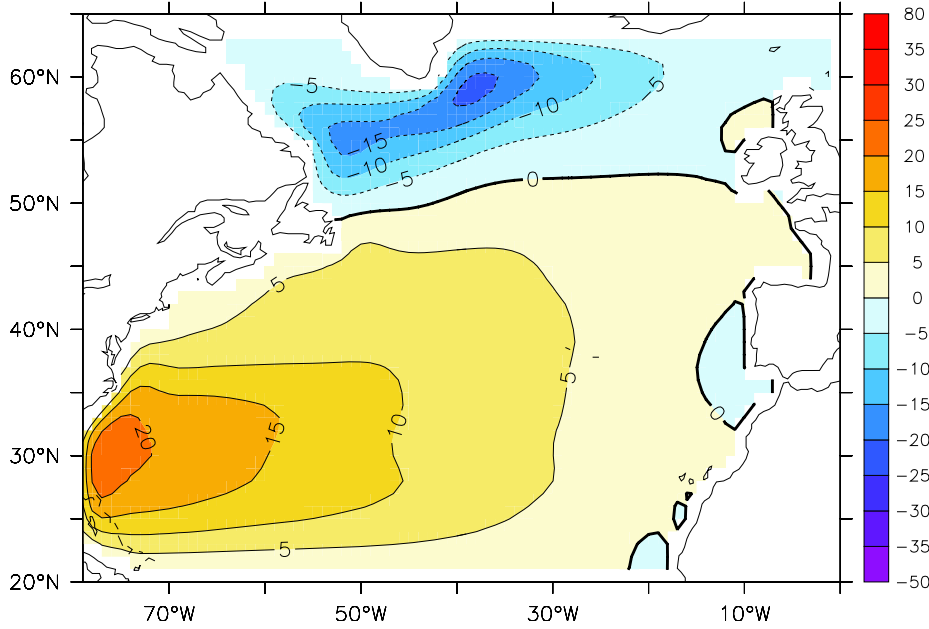


Fig. 14.6 The solution (streamfunction, in Sverdrups) to the Stommel-Munk problem numerically calculated for the North Atlantic, using the observed wind field. The model ocean has realistic geometry, but is flat-bottomed. The calculation qualitatively reproduces the large-scale patterns, including the subtropical and subpolar gyre and the western intensification of both, although the separation of the Gulf Stream from the coast is a little too far North. Compare with Fig. 14.2.⁹

14.3 ZONAL BOUNDARY LAYERS

The canonical wind stress [$\tau^x = -\tau_0 \cos(\pi y/a)$] is special because its curl vanishes at $y = 0, a$, and so the interior solution satisfies $\psi_I = 0$ at $y = 0, 1$. We cannot expect the real wind to be so accommodating. Consider, then, the problem forced by the linear wind profile,

$$\tau^x = \frac{\tau_0}{a} \left(y - \frac{a}{2} \right). \quad (14.50)$$

Scaling the variables in the usual way leads to the nondimensional problem

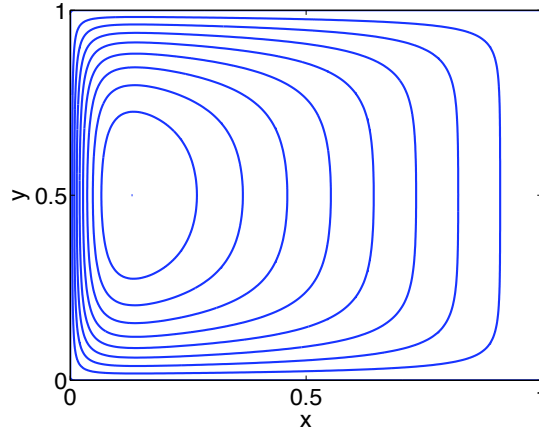
$$\epsilon_S \nabla^2 \hat{\psi} + \frac{\partial \hat{\psi}}{\partial \hat{x}} = -1. \quad (14.51)$$

The interior flow obeys $\partial \hat{\psi} / \partial \hat{x} = -1$ everywhere, and evidently does not satisfy $\hat{\psi} = 0$ at either $\hat{y} = 0$ or $\hat{y} = 1$; thus, boundary layers are needed at the meridional and zonal boundaries, and we let the solution be the sum of five parts,

$$\hat{\psi} = \psi_I + \phi_W + \phi_E + \phi_N + \phi_S, \quad (14.52)$$

with self-explanatory notation. The interior solution, ψ_I , is the solution of $\partial \psi_I / \partial \hat{x} = -1$, so that a solution satisfying $\psi_I = 0$ at $\hat{x} = 1$ is $\psi_I = (1 - \hat{x})$ and there is no need

Figure 14.7 Solutions to the Stommel problem with a windstress that increases linearly from $y = 0$ to $y = 1$, as in (14.50). The interior solution is $\psi_I = (1 - x)$, or $v_I = -1$, necessitating zonal boundary layers at $y = 0$ and $y = 1$, as well as a western boundary layer at $x = 0$.



for an eastern boundary layer. By the same methods we used in section 14.1.3 the western boundary layer correction is easily found to be

$$\phi_W = -e^{-\hat{x}/\epsilon_S}. \quad (14.53)$$

It remains to find ϕ_N and ϕ_S , the boundary layer corrections at the northern and southern boundaries.

Consider the boundary layer correction at $y = 1$, and introduce the stretched coordinate α where $\epsilon' \alpha = y - 1$ and where ϵ' is a small parameter. Thus, we let $\phi_N = \phi_N(x, \alpha)$, and on substituting (14.52) into (14.51) we find

$$\epsilon_S \left(\nabla^2 \psi_I + \frac{\partial^2 \phi_N}{\partial \hat{x}^2} + \frac{1}{\epsilon'^2} \frac{\partial^2 \phi_N}{\partial \alpha^2} \right) + \frac{\partial \psi_I}{\partial \hat{x}} + \frac{\partial \phi_N}{\partial \hat{x}} = -1, \quad (14.54)$$

having neglected the small contributions from the other boundary layer streamfunctions (such as ϕ_S). To obtain a nontrivial balance we choose $\epsilon'^2 = \epsilon_S$ and obtain the dominant balance

$$\frac{\partial^2 \phi_N}{\partial \alpha^2} + \frac{\partial \phi_N}{\partial \hat{x}} = 0. \quad (14.55)$$

The boundary conditions necessary to complete the solution are:

- (i) The total streamfunction (interior plus boundary correction) must vanish at the northern boundary. That is, $\phi_N(\hat{x}, \alpha = 0) = -\psi_I(\hat{x}, \hat{y} = 1) = -(1 - \hat{x})$.
- (ii) The boundary solution should match to the interior streamfunction far from the boundary. That is $\phi_N(\hat{x}, \alpha \rightarrow \infty) = 0$.
- (iii) At the eastern boundary ϕ_N should also vanish, else it would provide a velocity into the eastern wall. Thus, $\phi_N(1, \alpha) = 0$.

Actually obtaining the solution is algebraically tedious, however. The solution at the southern boundary is obtained in an analogous way, and the complete solution is illustrated in Fig. 14.7. The nondimensional thickness of the zonal (northern and southern) boundary layers is $\epsilon_S^{1/2}$, which, because it scales like the half power of a small number, is much thicker than the western boundary current. The thickness arises from the dimensional equations that dominate at the boundary, namely

$$r \frac{\partial^2 \psi}{\partial y^2} + \beta \frac{\partial \psi}{\partial x} = 0. \quad (14.56)$$

This follows from (14.18) by noting that the Laplacian operator must be dominated by the derivatives in the y -direction, and the β term has an (interior) component that annihilates the wind-stress-curl, plus a boundary layer correction to balance the Laplacian. Inspection of (14.56) yields a dimensional thickness $L_Z \sim \sqrt{ra/\beta} = \epsilon_S^{1/2} a$, where a is the length-scale in the x -direction.

14.4 THE NONLINEAR PROBLEM

In the nonlinear problem we seek solutions to

$$\frac{\partial \zeta}{\partial t} + J(\psi, \zeta) + \beta \frac{\partial \psi}{\partial x} = -r \nabla^2 \psi + \text{curl}_z \tau_T + \nu \nabla^2 \zeta, \quad (14.57)$$

which we have written in dimensional form. In the Stommel problem we set $\nu = 0$ and in the Munk problem we set $r = 0$. In general solutions will be time-dependent and turbulent and this will create motion on small scales, so that ν cannot be neglected. The ‘steady nonlinear Stommel-Munk problem’ is

$$J(\psi, \zeta) + \beta \frac{\partial \psi}{\partial x} = -r \nabla^2 \psi + \text{curl}_z \tau_T + \nu \nabla^2 \zeta. \quad (14.58)$$

We can scale this by first supposing that the leading order balance is Sverdrupian (i.e., $\beta \partial \psi / \partial x \sim \text{curl}_z \tau_T$), from which we obtain the scales $\Psi = |\tau|/\beta$ and $U = |\tau|/(\beta L)$. Equation (14.58) may then be nondimensionalized to yield

$$R_\beta J(\hat{\psi}, \hat{\zeta}) + \frac{\partial \hat{\psi}}{\partial \hat{x}} = -\epsilon_S \nabla^2 \hat{\psi} + \text{curl}_z \hat{\tau}_T + \epsilon_M \nabla^2 \hat{\zeta} \quad (14.59)$$

where $R_\beta = U/\beta L^2 = |\tau|/(\beta^2 L^3)$, the β -Rossby number for this problem, is a measure of the nonlinearity. Evidently, the nonlinear term increases in importance with increasing wind-stress and for a smaller domain (see also problem 14.8).

14.4.1 A perturbative approach

A direct attack on the full nonlinear problem (14.58) is possible only through numerical methods, so first we shall explore the problem perturbatively, assuming the nonlinear term to be small; the analysis is straightforward, albeit messy. We begin with the Stommel problem, (14.59) with $\epsilon_M = 0$, and expand the streamfunction in terms of R_β ,

$$\hat{\psi} = \psi_0 + R_\beta \psi_1 + \dots \quad (14.60)$$

Now substitute this into (14.59) and equate powers of R_β . The lowest order problem is simply

$$\epsilon_S \nabla^2 \psi_0 + \frac{\partial \psi_0}{\partial \hat{x}} = \text{curl}_z \hat{\tau} \quad (14.61)$$

which is the Stommel problem we have already solved. At next order,

$$\epsilon_S \nabla^2 \psi_1 + \frac{\partial \psi_1}{\partial \hat{x}} = J(\psi_0, \zeta_0) \quad (14.62)$$

This equation has precisely the same form as the Stommel problem, with the known nonlinear term on the right playing the part of the wind-stress. The algebra to obtain the solution is rather tedious, because the right-hand side varies with both $\hat{x}$ and

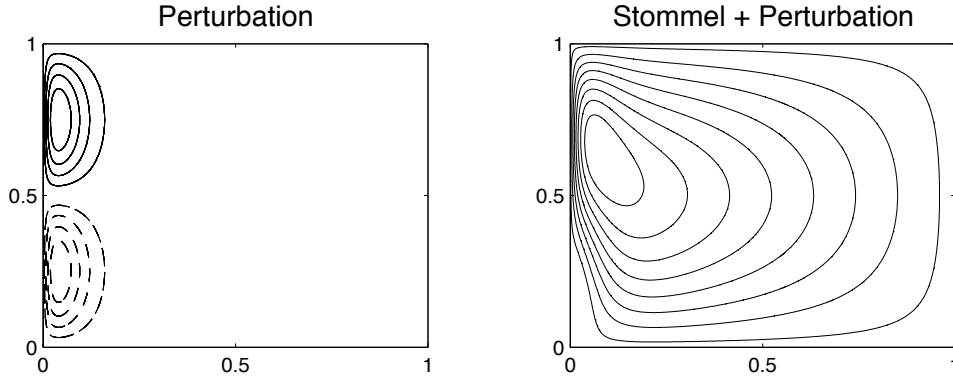


Fig. 14.8 The nonlinear perturbation solution of the Stommel problem, calculated according to (14.63). On the left is the perturbation, $-R_\beta \pi^3 / (2\epsilon_S^2) \sin(2\pi\hat{y}) \hat{x} e^{-\hat{x}/\epsilon_S}$, and on the right is the reconstituted solution, using $R_\beta = 10^{-4}$ and $\epsilon = 0.04$. Dashed contours are negative.

$\hat{y}$, but this is much ameliorated by the use of computer algebraic manipulation languages. For the canonical wind stress $\tau^x = -\tau_0 \cos(\pi\hat{y})$ the corrected solution, in the boundary layer approximation and ignoring any corrections at the zonal boundaries, is found to be:¹⁰

$$\hat{\psi} \approx \sin(\pi\hat{y})(1 - \hat{x} - e^{-\hat{x}/\epsilon_S}) - \frac{R_\beta \pi^3}{2\epsilon_S^3} \sin(2\pi\hat{y}) \hat{x} e^{-\hat{x}/\epsilon_S}. \quad (14.63)$$

The solution is illustrated in Fig. 14.8. The perturbation is antisymmetric about $\hat{y} = 1/2$, being positive for $\hat{y} > 1/2$ and negative for $\hat{y} < 1/2$. This tends to move the center of the gyre polewards, narrowing and intensifying the flow in the polewards half of the western boundary current, whereas the western boundary current equatorwards of $\hat{y} = 1/2$ is broadened and weakened. The net effect is that the center of the gyre is pushed polewards — essentially because the western boundary current is advecting the vorticity of the gyre polewards. In the perturbation solution the advection is both by and of the linear Stommel solution; thus, negative vorticity is advected polewards, intensifying the gyre in its polewards half, weakening it in its equatorward half. The solution illustrated in Fig. 14.8 has $\epsilon = 0.04$ and $R_\beta = 10^{-4}$; for larger values of R_β the perturbation itself starts to dominate.

The problem with this perturbative approach is that boundary layer solution to the Stommel problem does not calculate derivatives accurately, so that the nonlinear term $J(\psi, \nabla^2 \psi)$ is poorly approximated in the western boundary layer; however, in the interior where the errors are small the perturbative correction is negligible. A more accurate perturbative approach begins with the *exact* solution to the Stommel problem, and then proceeds in the same way. However, the analytic effort is considerable, and the intuitive sense of the way nonlinearities affect the solution is not apparent. (See also problem 14.6.)

14.4.2 A numerical approach

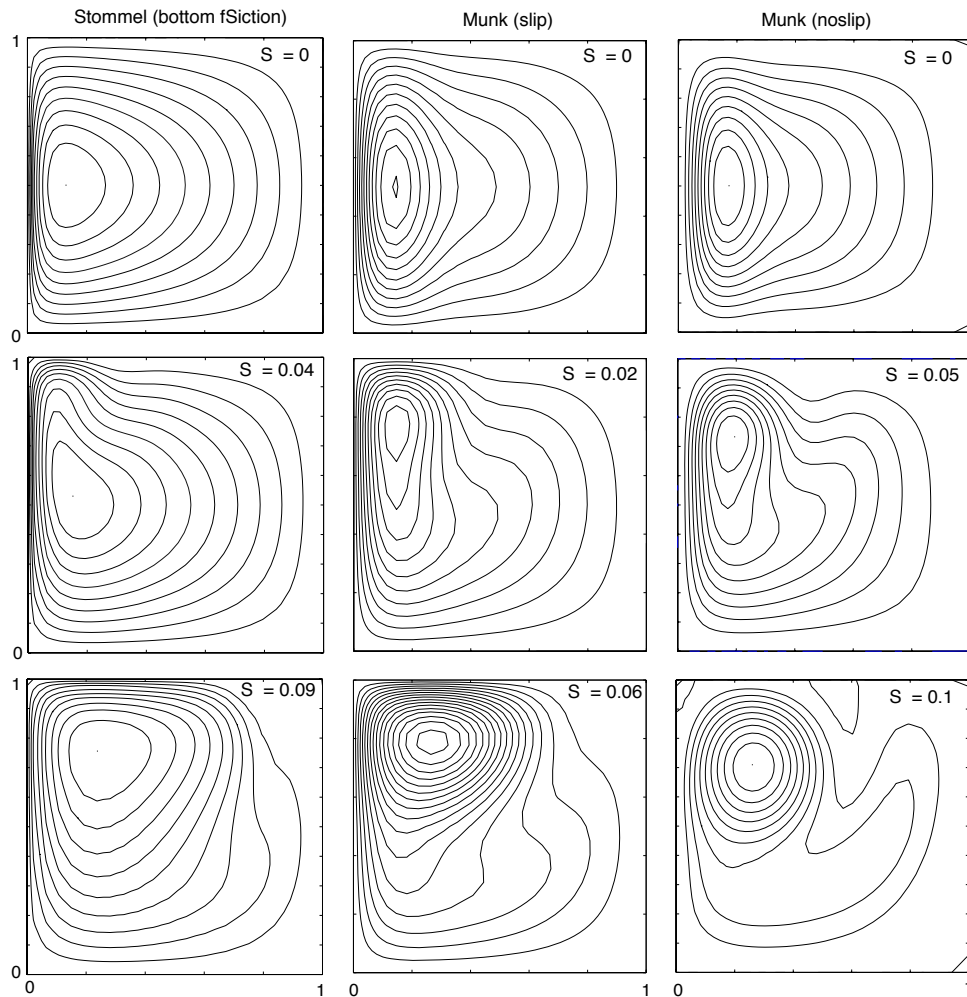


Fig. 14.9 Streamfunctions in solutions of the nonlinear Stommel and Munk problems, obtained numerically with a Newton's method, for various values of the nonlinearity parameter $S = R_\beta^{1/2}$. As in the perturbation solution, for small values of nonlinearity the center of the gyre moves polewards, strengthening the boundary current in the north-western quadrant (for a northern-hemisphere solution). As nonlinearity increases, the recirculation of the gyre dominates, and the solutions become increasingly inertial.

Fully nonlinear solutions show qualitatively similar effects to those seen in the perturbative solutions, as we see in Fig. 14.9, where the solution to (14.59) for the Stommel and Munk problems are obtained numerically by a Newton's method.¹¹ Just as with the perturbative procedure, for both the Stommel and Munk problems small values of nonlinearity lead to the poleward advection of the gyre's anticyclonic vorticity in the western boundary current, strengthening and intensifying the boundary current in the northwest corner. A higher level of nonlinearity results in a strong recirculating

regime in the upper westward quadrant, and ultimately much of the gyre's transport is confined to this regime. The western boundary current itself becomes less noticeable as nonlinearity increases, the more nonlinear solutions have a much greater degree of E-W symmetry than the linear ones, just as the fully nonlinear Fofonoff solutions (section 14.5.3).

Such qualitative effects do not depend on the precise formulation of the model, but the boundary conditions do play an important role in the detailed solution. For example, for a given value of R_β , nonlinearity has a stronger effect in the Munk problem with slip boundary conditions than with no-slip, because in the latter the velocity is reduced to zero at the boundary with a corresponding reduction in the advection term. However, these solutions themselves are unlikely to be relevant for larger values of nonlinearity, because then the flow becomes hydrodynamically unstable.

14.5 * INERTIAL SOLUTIONS

In this section we further explore inertial effects and ask: might purely inertial effects be sufficient to satisfy boundary conditions at the western boundary? Can we envision a purely inertial gyre circulation? Our question is motivated by the steady wind-driven, Rayleigh damped quasi-geostrophic equation, namely

$$J(\psi, \nabla^2 \psi) + \beta \frac{\partial \psi}{\partial x} = W_E - r \nabla^2 \psi \quad (14.64)$$

$$\frac{U^2}{L^2} \quad \beta U \quad |W_E| \quad r \frac{U}{L}, \quad (14.65)$$

where W_E is the wind forcing and the second line indicates the magnitude of the various terms. Since the inertial (advective) terms are higher order than the linear terms, indeed they are higher order than the Rayleigh drag, it is natural to wonder if they themselves might serve to satisfy the no-normal flow condition on the western boundary, without recourse to rather ill-defined frictional terms. The answer is no, as we see below, but nevertheless nonlinear effects may be important in the western boundary layer even if they are small in the interior.

14.5.1 Roles of friction and inertia

The need for friction

Consider the steady barotropic flow satisfying

$$\mathbf{u} \cdot \nabla q = \text{curl}_z \boldsymbol{\tau}_T + Fr \quad (14.66)$$

where $q = \nabla^2 \psi + \beta y$ and Fr represents frictional effects. Noting that $\mathbf{u}$ is divergence-free and integrating the left-hand side of (14.66) over the area, A , between two closed streamlines (ψ_1 and ψ_2 , say) and using the divergence theorem we find

$$\int_A \nabla \cdot (\mathbf{u}q) \, dA = \oint_{\psi_1} \mathbf{u}q \cdot \mathbf{n} \, dl - \oint_{\psi_2} \mathbf{u}q \cdot \mathbf{n} \, dl = 0. \quad (14.67)$$

Here, $\mathbf{n}$ is the unit vector normal to the streamline so that $\mathbf{u} \cdot \mathbf{n} = 0$. The integral of the wind-stress curl over the same area will not in general be zero. Now, we can take

these two streamlines as close together as we wish; thus, a balance in (14.66) can only be achieved if *every* closed contour passes through a region where frictional effects are non-zero. This does not mean that nonlinear terms may not locally dominate the friction, just that friction must be somewhere important. In the Stommel and Munk problems, it means that every streamline must pass through the frictional western boundary layer.

Frictional and inertial scales

As we have noted, the ratio of the size of the nonlinear terms to the linear terms is given by the beta Rossby number, $R_\beta = U/(\beta L^2)$. In the western boundary layer the length scales can be expected to be much smaller than the basin scale, and if the balance in the western boundary layer were between the nonlinear and beta term, as in

$$u \frac{\partial}{\partial x} \nabla^2 \psi \sim \beta \frac{\partial \psi}{\partial x}, \quad (14.68)$$

then the *inertial boundary layer thickness*, δ_I , would be given by

$$\delta_I = \left(\frac{U}{\beta} \right)^{1/2}. \quad (14.69)$$

This, of course, gives $R_\beta = 1$ if $L = \delta_I$. The more energetic the flow, the wider the region where nonlinearity is important, and the corresponding scale is sometimes called the Charney thickness.¹²

The linearized Stommel equation has a boundary layer of dimensional thickness of order $\delta_S = (r/\beta)$, obtained by equating $\beta \partial \psi / \partial x$ and $r \nabla^2 \psi$. This thickness is equal to the inertial boundary layer thickness when $U = (r^2/\beta)$. If $\delta_I > \delta_S$, i.e., if $U > (r^2/\beta)$, then nonlinearity *must* be important in the western boundary layer, because the nonlinear terms are at least as important as the beta term in (14.64). On the other hand, if the Stommel boundary layer is wider than the inertial boundary layer, there is no obvious need for nonlinearity to be important, since the Stommel boundary layer generates no length scales smaller than δ_I and R_β remains small.

14.5.2 Attempting an inertial western boundary solution

Although friction must be important, it is nevertheless instructive to try to find a purely inertial solution for the western boundary layer, and to see if and how the attempt fails. Let us suppose that the interior solution has purely zonal flow, with flow either towards or away from the western boundary current, and consider the dimensional equation of motion

$$J(\psi, \nabla^2 \psi + \beta y) = 0. \quad (14.70)$$

This has the general solution

$$\nabla^2 \psi + \beta y = G(\psi) \quad (14.71)$$

where G is an arbitrary function of its argument. Consider first the case with flow entering the boundary layer with a local velocity $-U$ (i.e., westward) — that is, $\psi = Uy$. The potential vorticity in this region (just outside our putative boundary layer)

is $Q_I = \beta y$. Thus, the relation between potential vorticity and streamfunction is $Q_I = \beta \psi_I / U$ and so

$$G(\psi) = \frac{\beta \psi}{U}. \quad (14.72)$$

Because we are assuming the flow is inviscid, the fluid will preserve this relationship between potential vorticity and streamfunction *even as it moves through the western boundary layer*. Thus, using (14.71) and (14.72), the flow in the interior *and* in the boundary layer are given by solutions of

$$\nabla^2 \psi - \frac{\beta \psi}{U} = -\beta y. \quad (14.73)$$

with $\psi = 0$ on the boundary. To obtain a solution, we let $\psi = \psi_I + \phi$, where $\psi_I = U y$ is the particular solution to (14.73) (and, of course, the interior flow). The boundary layer correction then obeys

$$\nabla^2 \phi - \frac{\beta \phi}{U} = 0. \quad (14.74)$$

with $\phi = -\psi_I$ at $x = 0$. Now, in the boundary layer, length-scales in the x -direction are much smaller than length-scales in the y -direction, and so (14.74) becomes approximately

$$\frac{\partial^2 \phi}{\partial x^2} - \frac{\beta \phi}{U} = 0. \quad (14.75)$$

(We could formalize this procedure by nondimensionalizing and introducing a stretched coordinate, as we did for the Stommel problem.) Solutions of (14.75) are $\phi = -\psi_I e^{-x/\delta_I}$, where $\delta_I = (U/\beta)^{1/2}$, and so the full solution is

$$\psi = \psi_I (1 - e^{-x/\delta_I}). \quad (14.76)$$

Clearly, this solution smoothly transitions into the interior solution for large x .

What about solutions exiting the boundary layer? We might attempt a similar procedure, but now the interior boundary condition that we must match is that of westward flow, namely $\psi_I = -U y$. Thus, analogously (14.73), to we seek solutions to the problem,

$$\nabla^2 \psi + \frac{\beta \psi}{U} = -\beta y \quad (14.77)$$

for which the homogeneous (boundary layer) equation is

$$\frac{\partial^2 \phi}{\partial x^2} + \frac{\beta \phi}{U} = 0. \quad (14.78)$$

Solutions of (14.78) are [c.f., (14.76)]

$$\psi = \psi_I (1 - e^{-x/\delta_I^*}) \quad (14.79)$$

where $\delta_I^* = i \delta_I = i (U/\beta)^{1/2}$. The solution is therefore wavelike, and does not transition smoothly to the interior flow. This suggests that friction must be important in allowing the boundary layer to smoothly connect to the interior for reasons connected to the conservation of potential vorticity, as discussed in the next subsection.¹³

In addition to this transition problem, we note that (14.76) and (14.79) together do not constitute a globally acceptable inviscid solution, for the simple reason that

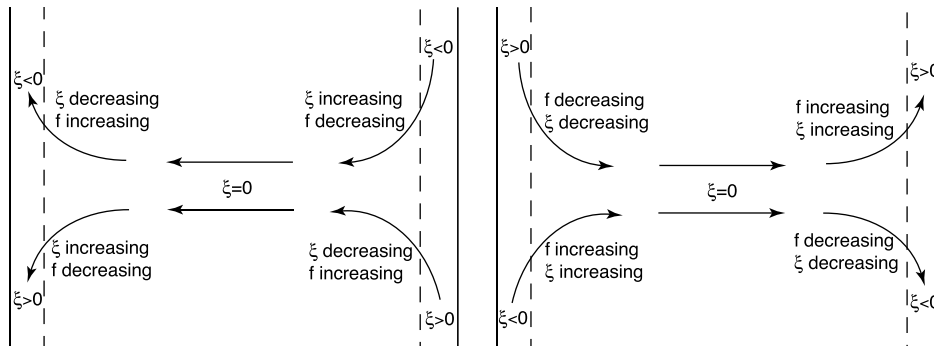


Fig. 14.10 Putative inertial boundary layers connected to a westward flowing internal flow (left panel) or eastward flowing internal flow (right panel), in the Northern Hemisphere. Westward flow into the western boundary layer, or flow emerging from an eastern boundary layer, is able to conserve its potential vorticity through a balance between changes in relative vorticity and Coriolis parameter. But flow cannot emerge smoothly from a western boundary layer into an eastward flowing interior and still conserve its potential vorticity. The right panel thus has inconsistent dynamics.

the flow in the westward flowing interior region has a different potential vorticity-streamfunction (q - ψ) relationship than does the eastward flowing interior. If these two regions are connected by a boundary layer, the flow in that boundary layer must be viscous or unsteady since the value of potential vorticity on the streamlines has changed, whereas inviscid flow conserves potential vorticity.

The connection between the boundary layer and the interior

We have seen that solutions with a boundary layer character that blend smoothly to a flow interior exist only for westward interior flow ($u_I = -U < 0$): an inertial western boundary layer on a beta-plane evidently cannot easily release fluid into the interior. The underlying reason for this stems from the conservation of potential vorticity, as we now explain (and see Fig. 14.10).

Conservation of potential vorticity demands that, in barotropic flow, $v_x - u_y + \beta y$ is a constant on streamlines. In the interior, relative vorticity, ζ , is zero, and in the meridional boundary layers it is effectively v_x . Consider first the case with westward flow in the interior (left panel of Fig. 14.10). Fluid from the irrotational interior approaches the western boundary where it is deflected either polewards or equatorwards. In the former case its relative vorticity falls, so allowing potential vorticity to be conserved because the reduction of ζ can be balanced by an increase in the planetary vorticity, f . Similarly, flow deflected southwards produces positive relative vorticity, which is compensated for by the reduced value of f . In the eastern boundary layer, southwards (northwards) moving flow has negative (positive) relative vorticity. As it emerges into the interior its relative vorticity increases (decreases), this being balanced by a fall (rise) in the value of f . Thus, we see that the solution with a westward flowing interior can indeed conserve potential vorticity, at both eastern and western boundaries.

On the other hand suppose the interior flow were eastward (right panel of Fig.

14.10). Flow moving polewards in the western boundary layer has negative relative vorticity. It cannot be freely released into an irrotational interior because f and ζ would then both need to increase, violating potential vorticity conservation. Flow moving southwards with positive relative vorticity similarly is trapped within the western boundary current, unless it meets a zonal boundary which allows a eastward moving boundary current with positive relative vorticity. Similar arguments show that an eastern boundary current cannot entrain fluid from an eastward flowing irrotational interior.

One might ask, why cannot we simply reverse the trajectory of all the fluid parcels in an inviscid flow and thereby obtain a solution with an eastward-flowing interior? The answer is that such a flow will only be a solution if we also reverse the direction of the earth's rotation, and so reverse the sign of β , and hence the location of the frictional boundary current.

14.5.3 A fully inertial approach: the Fofonoff model

Rather than attempt to match an inertial boundary layer with an interior Sverdrup flow, we may look for a purely inertial solution that holds basinwide, and such a construction is known as the *Fofonoff model*.¹⁴ That is, we seek global solutions to the inviscid, unforced problem,

$$J(\psi, \nabla^2 \psi + \beta y) = 0 \quad (14.80)$$

We should not regard this problem as representing even a very idealised wind-driven ocean; rather, we may hope to learn about the properties of purely inertial solutions, which might, in turn, tell us something about the ocean circulation.

The general solution to (14.80) is

$$\nabla^2 \psi + \beta y = Q(\psi) \quad (14.81)$$

where $Q(\psi)$ is an arbitrary function of its argument. For simplicity we choose the linear form,

$$Q(\psi) = A\psi + B, \quad (14.82)$$

where $A = \beta/U$ and $B = \beta y_0$, where U and y_0 are arbitrary constants. Thus, (14.81) becomes

$$\left(\nabla^2 - \frac{\beta}{U} \right) \psi = \beta(y_0 - y). \quad (14.83)$$

We will further choose $\beta/U > 0$, which we anticipate will provide a westward flowing interior flow, but which (from our experience in the previous section) is more likely to provide a meaningful solution than an eastward interior, and we will use boundary-layer methods to find a solution. A natural scaling for ψ is UL , where L is the domain size, and with this the nondimensional problem is

$$(\epsilon_F \nabla^2 - 1) \hat{\psi} = \hat{y}_0 - \hat{y}, \quad (14.84)$$

where $\epsilon_F = U/(\beta L^2)$, and if we take this to be small (note that $\epsilon_F = R_\beta$) we can find a solution by boundary-layer methods. As usual we write $\hat{\psi} = \psi_I + \phi$ where

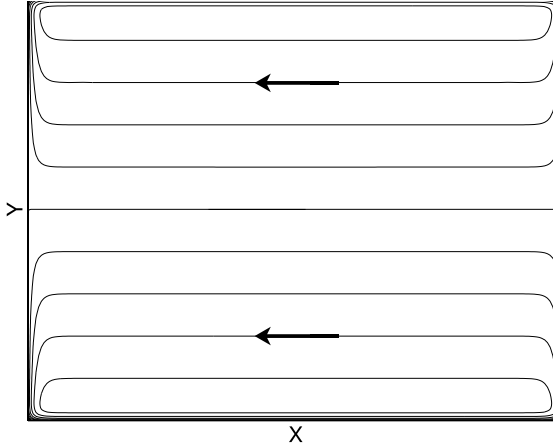


Figure 14.11 The Fofonoff solution. Plotted are contours (streamlines) of (14.88) in the plane $0 < x < x_E$, $0 < y < y_N$ with $U = 1$, $y_N = 1$, $x_E = y_N = 1$, $y_0 = 0.5$ and $\delta_I = 0.05$. The interior flow is westward everywhere, and $\psi = 0$ at $y = y_0$. In addition, boundary layers of thickness $\delta_I = \sqrt{U/\beta}$ bring the solution to zero at $x = (0, x_E)$ and $y = (0, y_N)$, excepting small regions at the corners.

$\psi_I = \hat{y} - \hat{y}_0$ [dimensionally, $U(y - y_0)$] and ϕ is the boundary layer correction, to be calculated separately for each boundary using

$$\epsilon_F \nabla^2 \phi - \phi = 0. \quad (14.85)$$

For example, at the northern boundary, $\hat{y} = \hat{y}_N$, the y -derivatives will dominate and (14.85) may be approximated by

$$\epsilon_F \frac{\partial^2 \phi}{\partial \hat{y}^2} - \phi = 0, \quad (14.86)$$

with solution

$$\phi = B e^{-(\hat{y}_N - \hat{y})/\epsilon_F}, \quad (14.87)$$

where $B = \hat{y}_0 - \hat{y}_N$, so satisfying the boundary condition that $\phi(\hat{x}, \hat{y}_N) = -\psi_I(\hat{x}, \hat{y}_N)$. We follow a similar procedure at the other boundaries to obtain the full solution, and in dimensional form this is

$$\psi = U(y - y_0) \left(1 - e^{-x/\delta_I} - e^{-(x_E - x)/\delta_I} \right) + U(y_0 - y_N) e^{-(y_N - y)/\delta_I} + U y_0 e^{-y/\delta_I}. \quad (14.88)$$

where $\delta_I = (U/\beta)^{1/2}$.

A typical solution is illustrated in Fig. 14.11. On approaching the western boundary layer, the interior flow bifurcates at $y = y_0$. The western boundary layer, of width δ_I , accelerates away from this point, being constantly fed by the interior flow. The westward return flow occurs in zonal boundary layers at the northern and southern edges, also of width δ_I . Flow along the eastern boundary layers is constantly being decelerated, because it is feeding the interior. If one of the zonal boundaries corresponds to y_0 (e.g., if $y_N = y_0$) there would be no boundary layer along it, since ψ is already zero at $y = y_0$. Rather, there would be westward flow along it, just as in the interior. Indeed, a slippery wall placed at $y = 0.5$ would have no effect on the solution illustrated in Fig. 14.11.

14.6 TOPOGRAPHIC EFFECTS ON WESTERN BOUNDARY CURRENTS

The above sections have emphasized the role of friction in satisfying the boundary conditions in the west. However, we should not think of friction as being the *cause* of the western boundary layer and in this section we shall show that if there are sloping sidewalls the role of friction is significantly different, and the western boundary current may even be largely inviscid!¹⁵ The key point is that the flow may be inviscid if it is able to follow potential vorticity contours. In a flat-bottomed western boundary layer the flow is moving to larger values of f (and not as a direct response to the wind) and so the flow *must* be frictional. However, if the sidewalls are sloping, then the flow may preserve its potential vorticity (essentially its value of f/h) if it moves offshore as it moves polewards. We will focus on homogeneous fluids, noting that the interaction of topography and stratification is a subtle and rather complex problem.

14.6.1 Homogeneous model

The potential vorticity evolution equation for a homogeneous model with topography and a rigid lid [c.f., (14.7)] may be written as:

$$\frac{Dq}{Dt} = \frac{F}{h}. \quad (14.89)$$

where $q = (\zeta + f)/h$ where $h = h(x, y)$ is the time-independent depth of the fluid, and F represents vorticity forcing and frictional terms. The advecting velocity is determined by noting that the mass conservation equation is just $\nabla \cdot [\mathbf{u}h(x, y)] = 0$, which allows us define the mass-transport streamfunction ψ such that

$$u = -\frac{1}{h} \frac{\partial \psi}{\partial y}, \quad v = \frac{1}{h} \frac{\partial \psi}{\partial x}. \quad (14.90)$$

The streamfunction is obtained from the vorticity by solving the elliptic equation,

$$\nabla \cdot \left(\frac{1}{h} \nabla \psi \right) = \zeta = qh - f. \quad (14.91)$$

The system — equations (14.89), (14.91) and (14.90) — is then closed. Unlike quasi-geostrophy, neither the Rossby number nor the topography need be small for the validity of the model. Including finite size topography but not stratification is not especially realistic *vis a vis* the real ocean, but nevertheless this model is physically realizable and a useful tool.

The ‘topographic Stommel problem’ is obtained by neglecting relative vorticity in the potential vorticity in (14.89) (i.e., let $q = f/h$) and using a linear drag acting on the vertically integrated fields for friction, and a windstress curl forcing. Multiplying (14.89) by h , expanding the advective term and omitting time dependence gives

$$J \left(\psi, \frac{f}{h} \right) = -r \nabla^2 \psi + \text{curl}_z(\boldsymbol{\tau}_W/h), \quad (14.92)$$

where the boundary conditions are $\psi = 0$ at the domain edges and $\boldsymbol{\tau}_W$ represents the wind-stress.

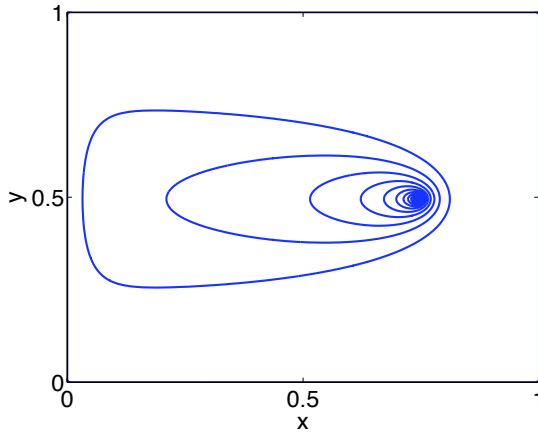


Figure 14.12 The β -plume, namely the Green's function for the Stommel problem. Specifically we plot the solution of (14.94) with $\psi = 0$ at the walls, and a delta-function source at $x = 0.75$, $y = 0.5$. The streamfunction trails westward from the source, as if it were a tracer being diffused while being advected westward along lines of constant f .

14.6.2 Advective dynamics

An illuminating way to begin to study the problem is to write (14.92) in the form¹⁶

$$J(\Psi, \psi) = +r\nabla^2\psi - \text{curl}_z(\boldsymbol{\tau}_W/h), \quad (14.93)$$

where $\Psi \equiv f/h$. Thus, noting that $J(\Psi, \psi) = \mathbf{U} \cdot \nabla \psi$ where $\mathbf{U} = \mathbf{k} \times \nabla(f/h)$, we regard ψ as being advected by the pseudo-velocity $\mathbf{U}$. This advection is along f/h contours and is quasi-westward, meaning that high values of potential vorticity lie to the right. Eq. (14.93) is then an advection-diffusion equation for the tracer ψ , with the advection occurring along f/h contours. The 'source' of ψ is the wind-stress curl, and ψ is diffused by the first term on the right-hand side of (14.93). This same interpretation applies to the original Stommel problem, of course, where the pseudo-velocity, $-\beta\mathbf{i}$, is purely westward, and it is useful to first revisit this problem.

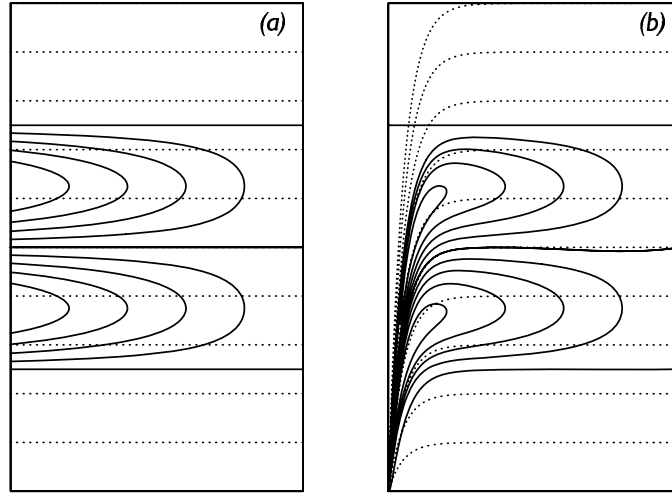
Consider, then, a flat-bottomed ocean, where the wind stress curl is just a point source at $\mathbf{x}_0$. With $\Psi = f$, (14.93) becomes

$$r\nabla^2\psi + \beta\frac{\partial\psi}{\partial x} = \delta(\mathbf{x} - \mathbf{x}_0). \quad (14.94)$$

This can be transformed to a Helmholtz equation by writing $\phi = \psi \exp(\beta x/2r)$, giving $\nabla^2\phi - [\beta/(2r)]^2\phi = \delta(\mathbf{x} - \mathbf{x}_0)$. This may then be solved exactly, and the solution (for ψ) is illustrated in Fig. 14.12 — this is the Green's function for the Stommel problem. The tracer ψ is 'advected' westwards along f/h contours — lines of latitude in this case — spreading diffusively as it goes; the resulting structure is called a β -plume. The western boundary layer results as a consequence of the f/h contours colliding with the western boundary, along with the need to satisfy the boundary condition $\psi = 0$. If there were no diffusion at all, ψ would just propagate westwards from the source, and if in addition the source were spatially distributed, for example as $\sin \pi y$, the solution streamfunction would represent Sverdrup interior flow.

Now consider the case with a sloping sidewall on the western boundary. The f/h contours (the dashed lines in Fig. 14.13b) tend to converge at the southwest corner of the domain, and only where f/h contours intersect the boundary is a diffusive boundary layer required. The solution to (14.94) with $r = 0$ with a conventional

Figure 14.13 The two-gyre Sverdrup flow (solid contours) for (a) a flat-bottomed domain and (b) a domain with sloping sidewalls. The f/h contours are dashed.¹⁷



‘two-gyre’ wind-stress are also illustrated in that figure; this is the Sverdrup flow for this problem and it evidently satisfies $\psi = 0$ on the western boundary. In terms of the interpretation above, wind-stress provides a source for the streamfunction ψ and the latter is advected *pseudo-westward* — i.e., along potential vorticity contours, with higher values to the right. The source in this case is distributed over the entire domain, but the contours all converge in the southwest corner. The (numerically obtained) solution to the associated Stommel problem is illustrated in Fig. 14.14, and the western boundary current in this case is no longer a frictional boundary layer. Friction is necessarily important where the flow crosses f/h contours; linear theory suggests that this will occur at the southwest corner. It also occurs on the western boundary where the f/h contours are densely packed and the vorticity in the topographic Sverdrup flow is large, and the friction enables the flow to move across the f/h contours. (In the flat bottomed case the f/h contours are zonal and friction allows the flow to move meridionally.)

14.6.3 Bottom pressure torques and form drag

In the homogeneous problem f and h appear only in the combination f/h , and we may solve the problem entirely without considering pressure effects. It is nevertheless informative to think about how the pressure interacts with the topography to produce meridional flow, and as way toward addressing the effects of stratification. The geostrophic momentum equation is

$$\mathbf{f} \times \mathbf{u} = -\nabla \phi + \mathbf{F} \quad (14.95)$$

where $\mathbf{F}$ represents both wind forcing and frictional terms. Integrating this over the depth of the ocean (with $z = 0$ at the top), and using Leibnitz’s rule ($\nabla \int_{\eta_B}^0 \phi \, dz = \int_{\eta_B}^0 \nabla \phi \, dz - \phi_B \nabla \eta_B$) to evaluate the pressure term, gives

$$\mathbf{f} \times \bar{\mathbf{u}} = -\nabla \bar{\phi} - \phi_B \nabla \eta_B + \bar{\mathbf{F}}. \quad (14.96)$$

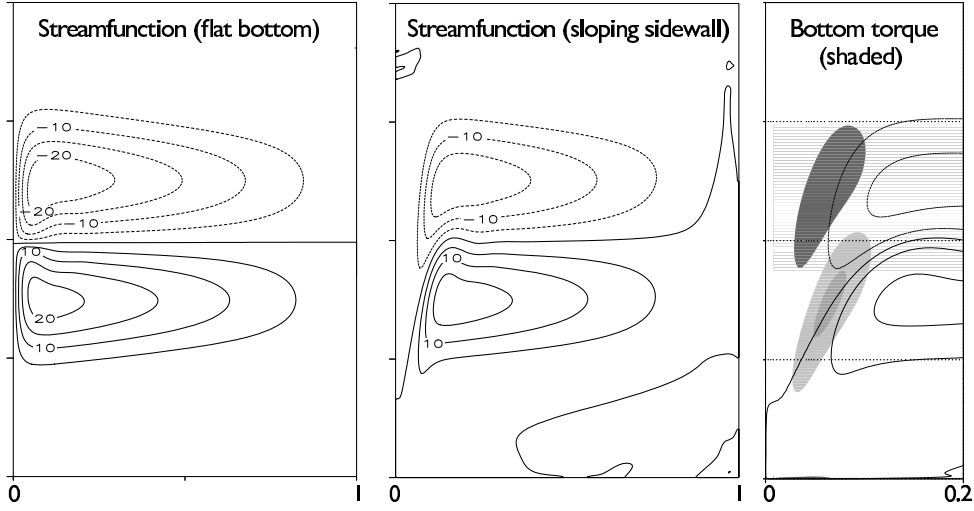


Fig. 14.14 The numerically obtained steady solution to the homogenous problem with a two-gyre forcing and friction, for a flat-bottomed domain and a domain with sloping western sidewall. The shaded regions in the right panel show the regions where bottom pressure torque is important, in the meridional flow of the western boundary currents.¹⁸

Here, the overbar denotes a vertical integral (e.g., $\bar{\mathbf{u}} = \int_{\eta_B}^0 \mathbf{u} dz$), ϕ_B is the pressure at $z = \eta_B$ and η_B is the z -coordinate of the bottom topography. We take the top of the ocean at $z = 0$, and note that $\nabla \eta_B = -\nabla h$ where h is the fluid thickness. Taking the curl of (14.96) gives

$$\beta \bar{v} = -J(\phi_B, \eta_B) + \text{curl}_z \bar{\mathbf{F}} \quad (14.97)$$

This equation holds for both a stratified and homogeneous fluid. The first term on the right-hand side is the *bottom pressure torque*, discussed more below.

Equation (14.97) is similar to (14.92), in that both arise from (14.96). To derive a equation with the same form as (14.92) but valid for a stratified fluid, we write the vertical integral of the pressure as

$$\int_{-h}^0 \phi dz = \int_{-h}^0 [d(\phi z) - z(\partial \phi / \partial z) dz] = \int_{-h}^0 [d(\phi z) - z b dz] = h \phi_B + \Gamma, \quad (14.98)$$

using hydrostatic balance, and where $\Gamma \equiv -\int_{-h}^0 z b dz$. Using (14.98) in (14.96) we obtain:

$$\mathbf{f} \times \bar{\mathbf{u}} = -h \nabla \phi_B + \nabla \Gamma + \bar{\mathbf{F}}. \quad (14.99)$$

The curl of this equation just gives back (14.97), but if we divide by h before taking the curl we obtain

$$J(\psi, f/h) + J(h^{-1}, \Gamma) = \text{curl}_z (\bar{\mathbf{F}}/h). \quad (14.100)$$

If the stratification vanishes (i.e., b is constant) then Γ is function of h alone and $J(h^{-1}, \Gamma) = 0$, and (14.100) reprises (14.92), given an appropriate choice of $\mathbf{F}$. The second term in (14.100) is known as the JEBAR term — joint effect of baroclinicity and relief — and it couples the depth integrated flow with the baroclinic flow.¹⁹ Thus, in

the bottom is not flat, the Stommel-Munk models are not solutions for the vertically integrated flow.

Bottom pressure torque in a homogeneous gyre

The first term on the right-hand side of (14.97) is the *bottom pressure torque*, T_B . It be written as the curl of a form stress, $\mathbf{\tau}_{form} = -\phi_B \nabla \eta_B$; that is, $T_B \mathbf{k} = \nabla_3 \times \mathbf{\tau}_{form} = -\nabla \phi_B \times \nabla \eta_B$, and so is non-zero whenever the pressure gradient has a component parallel to the topographic contour. Note that for unforced and inviscid flow we have both $J(\psi, f/h) = 0$ and, from (14.97), $\beta v = T_B$. That is, flow along f/h contours is equivalent to meridional flow balanced by a bottom pressure torque. On the other hand, if the domain is flat-bottomed, all meridional flow is forced or viscid.

However, unlike viscosity, the bottom pressure torque cannot balance the effects of the wind-stress curl when integrated over the whole domain, or indeed when integrated over an area bounded by a line of constant ϕ_B or constant η_B , because its integral over such an area vanishes and (14.92) cannot be balanced if $r = 0$. Locally, however, bottom pressure torque can be as or more important than either the wind forcing or friction. In the numerical simulations shown in Fig. 14.14, it is the bottom pressure torque term that largely balances the polewards flow (βv) in the vorticity equation in parts of western boundary current, with friction acting in the opposite sense. That is, over some regions where the flow is crossing f/h contours we have the balance

$$[\beta v] \approx [\text{Bottom pressure torque}] - [\text{Friction}], \quad (14.101)$$

where the terms in square brackets are positive, and friction is small. In contrast, in the flat-bottomed case in the western boundary layer we have the classical balance $[\beta v] \approx +[\text{Friction}]$, with both terms positive.

Now consider the balance of momentum, integrated zonally across the domain. We write the vorticity equation (14.97) in the form

$$\nabla \cdot (f \bar{\mathbf{v}}) = -\text{curl}_z(\phi_B \nabla \eta_B) + \text{curl}_z \mathbf{\tau}_T - \text{curl}_z \mathbf{\tau}_B, \quad (14.102)$$

where $\mathbf{\tau}_T$ is the wind stress at the top and $\mathbf{\tau}_B$ the frictional stress at the bottom. Integrate (14.102) over the area of a zonal strip bounded by two nearby lines of latitude, y_1 and y_2 , and the coastlines at either end. The term on the left-hand side vanishes by mass conservation and using Stokes's theorem we obtain:

$$\int_{y_1} \phi_B \frac{\partial \eta_B}{\partial x} dx - \int_{y_2} \phi_B \frac{\partial \eta_B}{\partial x} dx = \int_{y_1} (\tau_T^x - \tau_F^x) dx - \int_{y_2} (\tau_T^x - \tau_F^x) dx. \quad (14.103)$$

If the topography is non-zero, there is nothing in this equation to prevent the wind-stress being balanced by the form stress terms, with the friction being a negligible contribution. If, for example, friction were to be confined to the southwest corner, then bottom pressure torque is the proximate driver of fluid polewards in the western boundary current. This may hold only if the scale of the sloping sidewall is greater than the thickness of the Stommel layer; if the converse holds then the sidewalls appear to be essentially vertical to the flow. If the sidewalls are truly vertical, then the form stress is confined to delta-functions at the walls. Friction must then be important even in the zonal balance, because if we restrict the integral in (14.103) to

a strip that does not quite reach the sidewalls, the left-hand side vanishes identically and the wind-stress can only be balanced by friction.

To conclude this discussion, we note that the effects of topography are likely greater in homogeneous fluids than in stratified fluids, because the stratification may partially shield the wind-driven upper ocean from feeling the topography, but we leave the exploration of that topic for another day.

14.7 * VERTICAL STRUCTURE OF THE WIND-DRIVEN CIRCULATION

We now examine the vertical structure of the wind-driven flow.²⁰ We pose the problem using the quasi-geostrophic equations, taking the overall stratification of the ocean as a given. Of course the production of the stratification is itself one of the most important problems of physical oceanography, but we leave that to the next chapters.

14.7.1 A Two-Layer Quasi-Geostrophic Model

Scales of motion

We will concern ourselves with scales that are sufficiently larger than the deformation radius that we can ignore relative vorticity relative to vortex stretching and the β -effect. We must be careful in so doing, because quasi-geostrophic scaling applies only to scales that are not significantly larger than the deformation scale; thus, our analysis will be formally valid under the following set of inequalities:

$$\begin{aligned} \beta L &\ll f_0 && \text{(small variations in Coriolis parameter)} \\ \beta L &> U/L && \text{(to ignore relative vorticity compared to planetary vorticity)} \\ L^2 &> L_d^2 && \text{(to ignore relative vorticity compared to vortex stretching)} \\ Ro L^2 &\ll L_d^2 && \text{(to keep the variations in stratification small),} \end{aligned}$$

where L_d is the deformation radius and L the scale of the motion. Since in the mid-latitude ocean $L_d \approx 10^5$ m, then the above inequalities are reasonably well satisfied for $L \approx 10^6$ m and $U \approx 0.1$ m s⁻¹ with $\beta = 10^{-11}$ m⁻¹ s⁻¹ and $f_0 = 10^{-4}$ s⁻¹. The first and last of the inequalities are standard quasi-geostrophic requirements, with the ‘ $\gg$ ’ symbol denoting the asymptotic ordering. The middle two inequalities are taken within the quasi-geostrophic dynamics, and are needed in order to ignore relative vorticity and give a balance between the β -effect and vortex stretching. The simultaneous satisfaction of all these conditions may seem restrictive, but the plangent dynamics contained within the quasi-geostrophic equations and the generality of the method employed below will suggest that the principle results obtained may transcend the particular limitations of the equations used.

Constructing the model

We now make the following simplifications for our model ocean.

- (i) We use the two-layer quasi-geostrophic equations, with layers of equal thickness.
- (ii) We seek only statistically-steady solutions.
- (iii) We include a frictional term derived by assuming potential vorticity is fluxed downgradient. Given the neglect of relative vorticity, this is equivalent to an interfacial drag.

(iv) We will neglect the western boundary layer.

Because of the equal-layer-thickness assumption, which we make primarily for algebraic simplicity, it is best considered as model for the upper ocean above a level where the vertical velocity is approximately zero. The equations of motion are then

$$J(\psi_1, q_1) = \frac{1}{H_0} \text{curl}_z \boldsymbol{\tau}_T - \nabla \cdot \mathbf{T}_1, \quad J(\psi_2, q_2) = -\nabla \cdot \mathbf{T}_2 \quad (14.104a,b)$$

where

$$q_1 = \beta y + F(\psi_2 - \psi_1), \quad q_2 = \beta y + F(\psi_1 - \psi_2). \quad (14.105a,b)$$

Here, $F = f_0^2 / (g' H_0) = 1/L_d^2$ is a measure of the stratification, where H_0 is the thickness of either layer, and the $\nabla \cdot \mathbf{T}$ terms are represent interfacial eddy stresses, which we parameterize by by a downgradient flux of potential vorticity:

$$\mathbf{T}_1 = -\kappa \nabla q_1 = -R \nabla (\psi_2 - \psi_1), \quad \mathbf{T}_2 = -\kappa \nabla q_2 = -R \nabla (\psi_1 - \psi_2), \quad (14.106)$$

where R and κ are constants, and $R = \kappa F$. We will be interested in the limit of small R , or more specifically $UFL/R \gg 1$, which is a large Péclet number condition. (The Péclet number is similar to a Reynolds number, but with the diffusivity replacing the kinematic viscosity.) So first consider the case when R is identically zero. An *exact* solution to (14.104) has $\psi_2 = 0$, so that (14.104a) becomes $\beta \partial \psi_1 / \partial x = H_0^{-1} \text{curl}_z \boldsymbol{\tau}_T$, with solution

$$\psi_1 = -\frac{1}{H_0 \beta} \int_x^{x_E} \text{curl}_z \boldsymbol{\tau}_T dx. \quad (14.107)$$

That is, there is no flow in the lower layer, and the upper layer solution is given by Sverdrup balance. The solution satisfies $\psi_1 = 0$ at $x = x_E$ and, because $\psi_2 = 0$, the nonlinear term on the left-hand side of (14.104a) vanishes identically.

General Solution

We now construct the solution without assuming $\psi_2 = 0$. Although the equations are nonlinear, we obtain a linear equation for the lower-layer streamfunction by adding (14.104a) and (14.104b), giving

$$J(\psi_1, \beta y + F(\psi_2 - \psi_1)) + J(\psi_2, \beta y + F(\psi_1 - \psi_2)) = \frac{1}{H_0} \text{curl}_z \boldsymbol{\tau}_T. \quad (14.108)$$

The nonlinear terms cancel leaving

$$J(\bar{\psi}, \beta y) = \frac{1}{H_0} \text{curl}_z \boldsymbol{\tau}_T, \quad \text{where } \bar{\psi} = \psi_1 + \psi_2, \quad (14.109a,b)$$

with solution [c.f., (14.107)]

$$\bar{\psi} = -\frac{1}{H_0 \beta} \int_x^{x_E} \text{curl}_z \boldsymbol{\tau}_T dx'. \quad (14.110)$$

This simply says that the vertically integrated flow obeys Sverdrup balance. For the canonical windstress

$$\boldsymbol{\tau}_T = -\tau_0 \cos \pi y \mathbf{i} \quad (14.111)$$

we obtain $\bar{\psi} = (\pi \tau_0 / \beta H_0) (x_E - x) \sin \pi y$. It is useful to define

$$\bar{q} \equiv (\beta y + F \bar{\psi}), \quad (14.112)$$

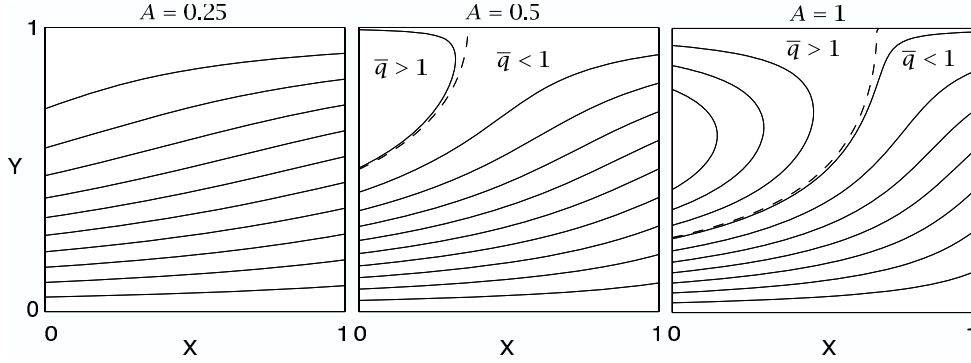


Fig. 14.15 Contours of $\bar{q} = \beta y + A \sin \pi y(1 - x)$, with $\beta = 1$. The dashed line is $\bar{q} = 1$, which separates the blocked region to the east ($\bar{q} < 1$) from the closed region to the west ($\bar{q} > 1$). See Fig. 14.16 for plots of the other fields.

and then $\bar{q} = \beta[y + A(1 - x) \sin \pi y]$, where $A = \pi \tau_0 / (\beta H_0)$ parameterizes the wind strength, and this is plotted in Fig. 14.15. For $\bar{q} < 1$ (below and to the right of the dashed line) all the geostrophic contours intersect the eastern boundary and the flow is ‘blocked’. For $\bar{q} > 1$ the flow is ‘closed’.

Lower layer

Although the full equations are nonlinear, using (14.112) we can obtain a linear equation for the lower layer. Because the Jacobian of a field with itself vanishes, (14.104b) and (14.105b) imply that

$$J(\psi_2, \bar{q}) = -\nabla \cdot \mathbf{T}_2, \quad (14.113)$$

and this is useful because $\bar{q}$ is a function of the wind, using (14.112). If $\nabla \cdot \mathbf{T}_2 = 0$ then

$$J(\psi_2, \bar{q}) = 0. \quad (14.114)$$

As well as the possibility that $\psi_2 = 0$ we now have the more general solution

$$\psi_2 = G(\bar{q}) \quad (14.115)$$

where G is an *arbitrary* function of its argument. Isolines of ψ_2 and $\bar{q}$ are then coincident. (Contours that are isolines of both streamfunction and potential vorticity are generally known as geostrophic contours.)

Consider a blocked isoline of $\bar{q}$; that is, one that intersects the eastern boundary (see again Fig. 14.15). The ψ_2 contour coincident with this has value zero at the eastern boundary (by the no-normal flow condition). Thus $\psi_2 = 0$ *everywhere* in the blocked region, and $q_2 = \bar{q}$. In this region the Sverdrup transport is carried everywhere by the upper layer, and the lower layer is at rest. This region is called a ‘shadow zone’, for the fluid is in the shadow of the eastern boundary, and it will re-appear in a model of the thermocline in chapter 16. In the region of closed contours, ψ_2 cannot be given by this argument. But if R is small, we can expect (14.114) to approximately hold, and that the presence of a small amount of dissipation will determine the functional relationship between ψ_2 and $\bar{q}$. Thus, in summary, there are two regions of flow:

- (i) The blocked region in which $\psi_1 \approx \bar{\psi} \gg \psi_2$ and ψ_1 is approximately given by (14.107).
- (ii) A closed region in which $\psi_2 = G(\bar{q}) + \mathcal{O}(R)$.

14.7.2 The form of $G(\bar{q})$

A general argument

In chapter 10 we showed that, within a region of closed contours, the values of a tracer that is materially conserved except for the effects of a small diffusion would become *homogeneous*. In the case at hand, potential vorticity is that tracer so that within potential vorticity contours or closed streamlines potential vorticity will become homogenized. If we can determine the value of q_2 within the region of closed contours, then from (14.105) ψ_2 is given by

$$\psi_2 = (1/2F)(\bar{q} - q_2) \quad (14.116)$$

and the solution would be complete. Now, outside the closed region $\psi_2 \ll \psi_1$, so that the outermost contour of the closed region must be characterized by $q_2 \approx \bar{q}$, for this makes ψ_2 continuous between closed and blocked regions. Thus, the value of q_2 within the closed homogeneous region is that of $\bar{q}$ (i.e., $\beta y + F\bar{\psi}$) on its boundary. Since this contour intersects the poleward edge of the domain, where $\bar{\psi}$ is zero, the value of this contour is just βy at $y = L$; that is, βL . Thus, within the closed region,

$$q_2 = \beta L. \quad (14.117)$$

A specific calculation

Now consider the steady, lower-layer potential vorticity equation (14.104b); since $J(\psi_2, F(\psi_1 - \psi_2)) = J(\psi_2, F(\psi_1 + \psi_2))$, (14.104b) may be written as

$$J(\psi_2, \bar{q}) = -\nabla \cdot \mathbf{T}_2 \quad (14.118)$$

Integrating around a closed contour of $\bar{q}$ the left-hand side vanishes and

$$R \int (\nabla \psi_1 - \nabla \psi_2) \cdot \mathbf{n} \, dl = 0 \quad (14.119)$$

or

$$\oint \mathbf{u}_1 \cdot d\mathbf{l} = \oint \mathbf{u}_2 \cdot d\mathbf{l} \quad (14.120)$$

thus, the deep circulation around a mean geostrophic contour (i.e., isoline of $\bar{q}$) is equal to the upper level circulation.

Now, previously we argued that

$$\psi_2 = G(\bar{q}) = G(\beta y + F(\psi_1 + \psi_2)). \quad (14.121)$$

where G is an arbitrary function of its argument. In order to satisfy (14.120) (a linear relation between $\mathbf{u}_1$ and $\mathbf{u}_2$), G must be a linear function, and so we write

$$\psi_2 = C \left[\frac{\beta y}{F} + (\psi_1 + \psi_2) \right] + B, \quad (14.122)$$

Summary of Wind-driven, Two-layer Solution

The vertically integrated flow in a wind-driven two-layer quasi-geostrophic model is determined by Sverdrup balance. The effects of eddies may be crudely parameterized by a downgradient diffusion of potential vorticity. If this is identically zero, then the lower layer flow is identically zero and the upper layer flow carries all the transport. If the diffusion is small but non-zero, the lower layer streamfunction approximately satisfies $J(\psi_2, \bar{q})$, where $\bar{q}$ is given by (14.112), and therefore ψ_2 is a function of $\bar{q}$ — that is, $\psi_2 \approx G(\bar{q})$. For a typical subtropical wind, contours of $\bar{q}$, and therefore contours of ψ_2 , are naturally divided into two regions (Fig. 14.15):

- (i) A blocked region (the shadow zone), in which contours of $\bar{q}$ intersect the eastern boundary, the lower layer flow is zero and the upper layer carries all the Sverdrup transport.
- (ii) A closed region in which (if we envision a nearly inviscid western boundary current) the flow recirculates. In this region we posit that the lower layer potential vorticity becomes homogeneous, with a value determined by the value at the region's boundary, and this in turn is determined by tracing $\bar{q}$ back to the domain boundary.

To satisfy a circulation constraint the function $G(\bar{q})$ must be a linear function, and given this the entire solution may be determined. If, for example, the wind is zonal and a function of y only, and $\text{curl}_z \tau_T = g(y)$ then, in both regions:

$$\bar{\psi} \equiv \psi_1 + \psi_2 = -\frac{1}{\beta H_0} g(y)(x_E - x), \quad \bar{q} \equiv \beta y + F\bar{\psi} \quad (\text{OC.1a,b})$$

In the blocked region:

$$\psi_2 = 0, \quad \psi_1 = -\frac{1}{\beta H_0} (x_E - x) g(y), \quad (\text{OC.2a})$$

$$q_1 = \beta y + F(\psi_1 - \psi_2), \quad q_2 = \beta y + F(\psi_2 - \psi_1). \quad (\text{OC.2b})$$

In the closed region:

$$q_2 = \beta L \quad (\text{by homogenization}), \quad (\text{OC.3a})$$

$$\psi_2 = \frac{1}{2F} (\bar{q} - \beta L), \quad \psi_1 = \bar{\psi} - \psi_2, \quad (\text{OC.3b})$$

$$q_1 = \beta y + F(\psi_2 - \psi_1) = 2\beta y - \beta L. \quad (\text{OC.3c})$$

For $g(y) = -\sin \pi y$ these solutions are illustrated in Fig. 14.15 and Fig. 14.16.

This approach provides a solution to the conundrum of what drives the lower ocean ocean, for if there are no eddy effects at all [i.e., $T_1 = T_2 = 0$ in (14.106)], then the lower layer flow is stationary. This seems unrealistic, for the upper layer flow could be made quite shallow. Another solution to this issue is provided in section 16.4, wherein it is assumed that the lower layers may outcrop and so feel the wind directly.

where C and B are constants. This may be re-arranged to give

$$\psi_1 = -C \frac{\beta y}{F} + (\psi_1 + \psi_2)(1 - C) - B. \quad (14.123)$$

This is consistent with (14.120) if $C = 1/2$. With this, (14.122) gives

$$\bar{q} = 2F(\psi_2 - B) \quad (14.124)$$

and the potential vorticity in the closed contour region of the lower layer is

$$q_2 = \beta y + F\bar{\psi} - 2F\psi_2 = -2FB. \quad (14.125)$$

That is, it is constant. Outside the closed contours $\psi_2 \ll \psi_1$ so that $q_2 \approx \bar{q} = \beta y + F\psi_1$. If we trace this contour to the edge of the domain where $\psi_1 = 0$ and $y = L$ then we see that the value of $\bar{q}$ on the contour, and hence q_2 in the closed region, is βL , as in (14.117), and $B = -\beta L/(2F)$. Using (14.124) then gives

$$\psi_2 = (2F)^{-1}(\bar{q} - \beta L). \quad (14.126)$$

Given ψ_2 and q_2 [from (14.117)], we obtain q_1 and ψ_1 using (14.105) and (14.109b), giving

$$q_1 = 2\beta y - \beta L, \quad \psi_1 = \bar{\psi} - \psi_2. \quad (14.127)$$

All these fields are illustrated in Fig. 14.16, and see the shaded box on the preceding page for a summary.

14.8 * A MODEL WITH CONTINUOUS STRATIFICATION

We now look the dynamics of the continuously stratified circulation, largely by way of an extension of our two-layer procedure. Let us first consider how deep the wind's influence is.

14.8.1 Depth of the Wind's Influence

The thermal wind-relationship in the form $f\partial u/\partial z = \partial b/\partial y$ implies a vertical scale H given by

$$H = \frac{fUL}{\Delta b} \quad (14.128)$$

where Δb is a typical magnitude of the horizontal variation of the buoyancy. We can relate this to the Ekman pumping velocity using the linear geostrophic vorticity equation, $\beta v = f\partial w/\partial z$, which, (assuming that the horizontal components of velocity are roughly similar, i.e., $V = U$), implies that

$$U = \frac{fW_E}{\beta H}, \quad (14.129)$$

Equations (14.128) and (14.129) may be combined to give an estimate of the depth of the wind-driven circulation, namely

$$H = \left(\frac{f^2 W_E L}{\beta \Delta b} \right)^{1/2}, \quad (14.130)$$

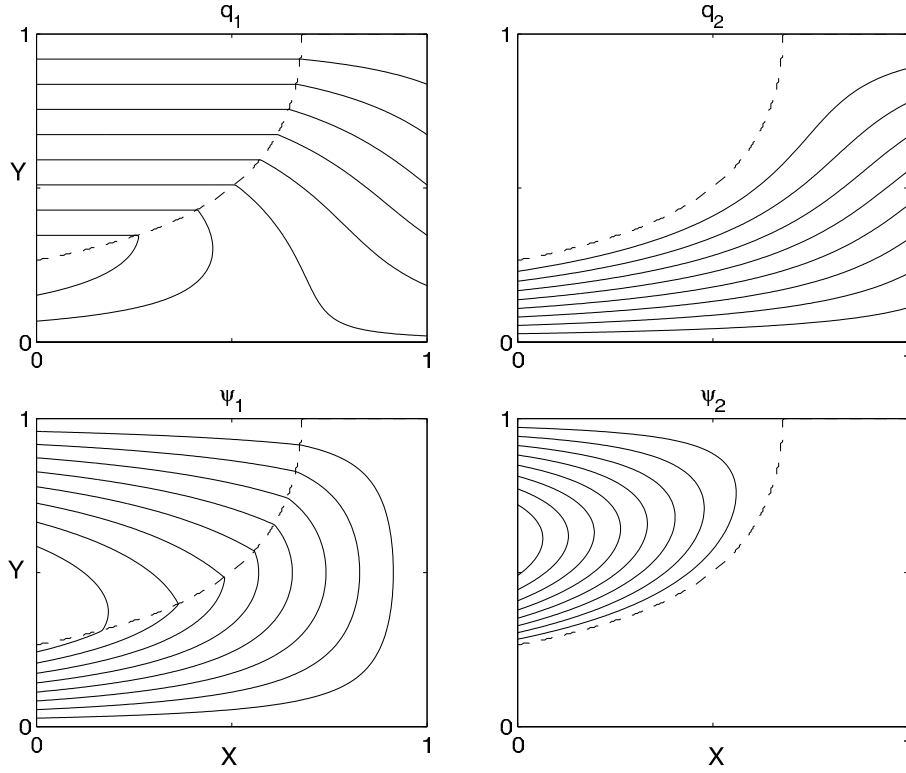


Fig. 14.16 Upper and lower level potential vorticity and streamfunction for the canonical windstress (14.111). The field of $\bar{q}$ is that of Fig. 14.15 with $A = 1$. The dashed line divides the blocked region from the closed region. The lower layer streamfunction ψ_2 is non-zero only in the closed region, and here $q_2 = \beta L$ and $q_1 = 2\beta y - \beta L$. In the blocked region the upper layer carries all of the Sverdrup transport. Both streamfunction and potential vorticity are continuous at the divide: $\psi_2 = 0$ and $q_2 = \bar{q} = \beta L$.

where L may be interpreted as the gyre scale. We now use quasi-geostrophic scaling to relate the horizontal temperature gradient to the stratification using the thermodynamic equation,

$$\frac{Db}{Dt} + wN^2 = 0, \quad (14.131)$$

with implied scaling

$$\Delta b = \frac{W_E N^2 L}{U} = \frac{N^2 \beta H L}{f_0}, \quad (14.132)$$

where the second equality uses (14.129). Using (14.130) and (14.132) gives

$$H = \left(\frac{W_E f^3}{\beta^2 N^2} \right)^{1/3}. \quad (14.133)$$

Potential vorticity interpretation

The estimate (14.133) can be obtained and interpreted more directly: *the wind-driven circulation penetrates as far as it can alter the potential vorticity q from its planetary value βy* . Recall that, ignoring relative vorticity,

$$q = \beta y + \frac{\partial}{\partial z} \left(\frac{f_0^2}{N^2} \frac{\partial \psi}{\partial z} \right). \quad (14.134)$$

The two terms are comparable if

$$\frac{f_0^2}{N^2 H^2} UL \approx \beta L \quad (14.135)$$

or

$$H^2 \approx \frac{f_0^2 U}{N^2 \beta}. \quad (14.136)$$

Using (14.129) to eliminate U in favour W_E recovers (14.133). Thus, for a given stratification, we have an estimate of the depth of the wind-driven circulation, or at least a scaling for depth of the vertical influence of the wind.

14.8.2 The complete solution

Armed with an estimate for the depth of the wind's influence, we can obtain a solution for the continuously stratified case analogous to that found in the two-layer case section 14.7. Our assumptions are:

- (i) In the limit of small dissipation streamfunction and potential vorticity have a functional relationship with each other.
- (ii) Potential vorticity is homogenized within closed isolines of q or ψ . The value of q within the homogenized pool is that of the outermost contour, which here is the value of q at the poleward edge of the barotropic gyre.
- (iii) Outside of the pool region, (i.e., below the depth of the wind's influence) the streamfunction is zero, and the potential vorticity is given by the planetary value, i.e., βy .

Given these, finding a solution is not difficult. If N^2 is constant and neglecting relative vorticity, the expression for potential vorticity is

$$q = \frac{\partial^2}{\partial z^2} \left(\frac{f_0^2}{N^2} \psi \right) + \beta y \quad (14.137)$$

We nondimensionalize by writing

$$z = \left(\frac{f_0^2 U}{N^2 \beta} \right)^{1/2} \hat{z}, \quad q = \beta L \hat{q}, \quad \psi = \hat{\psi} UL, \quad y = L \hat{y}, \quad w = \frac{U^2 f_0}{N^2 H} \hat{w}, \quad (14.138)$$

where the hatted variables are nondimensional, and the scaling for w arises from the thermodynamic equation $N^2 w \sim J(\psi, f_0 \psi_z)$. With this, (14.137) becomes

$$\hat{q} = \frac{\partial^2 \hat{\psi}}{\partial \hat{z}^2} + \hat{y}, \quad (14.139)$$

The flow is then given by solving the following equations:

$$\psi_{zz} + \gamma = \gamma_0, \quad -D(x, \gamma) < z < 0, \quad (14.140a)$$

$$\psi = 0, \quad z \leq -D(x, \gamma), \quad (14.140b)$$

where D is the (to-be-determined) depth of the bowl, γ_0 is a constant, and we have dropped the hats over the nondimensional variables. The solution in $D(x, \gamma) < z < 0$ corresponds to the closed region of the two-layer model, and the solution $z \leq -D(x, \gamma)$ corresponds to the blocked region of zero lower-layer flow. The constant γ_0 is the nondimensional value of potential vorticity within the pool region, and following our reasoning in the two-layer case this is the value of the potential vorticity at the northern boundary. Dimensionally this is βL , so that in nondimensional units $\gamma_0 = 1$.

The lower boundary conditions on (14.140a) is that $\psi = \psi_z = 0$ at $z = -D$, because in the abyss $\psi = \partial\psi/\partial z = 0$ and we require that both ψ and $\partial\psi/\partial z$ be continuous (note that the buoyancy perturbation is proportional to $\partial\psi/\partial z$). The solution that satisfies this

$$\psi = \frac{1}{2}(z + D)^2(\gamma_0 - \gamma), \quad (14.141)$$

and $\psi = 0$ for $z < -D$. To obtain an expression for D we first note that the nondimensional vertical velocity at $z = 0$ is given by

$$w = -J(\psi, \psi_z), \quad (14.142)$$

which, using (14.141), gives

$$w = \frac{1}{2}(z + D)^2(\gamma_0 - \gamma) \frac{\partial D}{\partial x}. \quad (14.143)$$

At $z = 0$ the vertical velocity is the Ekman pumping velocity and (14.143) becomes

$$D^2 \frac{\partial D}{\partial x} = \frac{2w_E}{(\gamma_0 - \gamma)} \quad (14.144)$$

But the Ekman pumping velocity is related to the barotropic streamfunction, ψ_B , by the Sverdrup relationship, so that integrating (14.144) gives

$$\boxed{D^3 = \frac{6\psi_B}{(\gamma_0 - \gamma)} = -\frac{6(x_E - x)w_E}{(\gamma_0 - \gamma)}}. \quad (14.145)$$

where the second equality holds if w_E is not a function of x . This is a solution for the depth of moving region, the bowl in which potential vorticity is homogenized. An expression for the streamfunction is then obtained by using (14.145) in (14.141), and is found to be

$$\psi = \begin{cases} \frac{1}{2} [z(\gamma_0 - \gamma)^{1/2} + (6\psi_B)^{1/3}(\gamma_0 - \gamma)^{1/6}]^2 & -D < z < 0, \\ 0 & z < -D. \end{cases} \quad (14.146)$$

The potential vorticity corresponding to this solution is

$$q = \begin{cases} \gamma_0 & -D < z < 0, \\ \gamma & z < -D. \end{cases} \quad (14.147)$$

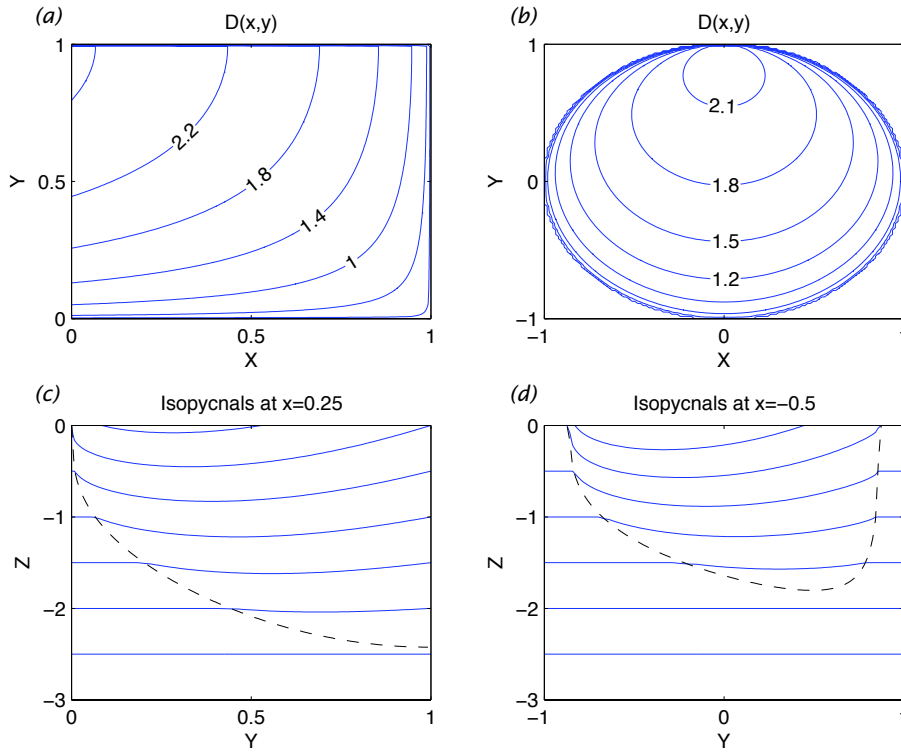


Fig. 14.17 Solutions of (14.112) for two different barotropic streamfunctions. On the left $\psi_B = (1-x) \sin \pi y$ and on the right $\psi_B = 1 - (x^2 + y^2)$ for $x^2 + y^2 < 1$, zero elsewhere. The upper panels show contours of the depth of the wind-influenced region (solutions of (14.145)). The depth increases to the northwest in the left panel, and to the north in the right panel, so that in both cases the area of the bowl shrinks with depth. The lower panels are contours of $z + (\beta L / f_0) \psi_z / 2$, with $\beta L / f_0 = 1/2$, obtained from (14.141) or (14.146), at $x = 0.25$ and $x = -0.5$ in the two cases. These are isopycnal surfaces, with a rather large value of $\beta L / f_0$ to exaggerate the displacement in the bowl region. The dashed lines indicate the boundary of the bowl region, outside of which the isopycnals are flat.

Solutions are illustrated in Fig. 14.17 and Fig. 14.18 for two different barotropic streamfunctions.

It is possible to extend models such as the one described above by appending a western boundary layer, and indeed the homogenization of potential vorticity depends upon the presence of such layer in allowing the flow to recirculation. However, as we saw in section 14.5.3, it is difficult for flow to leave a western boundary layer without the help of friction, and a neutrally stable, damped, stationary Rossby wave typically forms. The critical issue then is whether the presence of dissipation in the western boundary layer affects the homogenization of potential vorticity in the gyre itself. Numerical solutions with a quasi-geostrophic model do show that potential vorticity is able to homogenize under such circumstances. Homogenization in the real ocean has been observed in the Pacific and, somewhat less compellingly, in the Atlantic.²¹

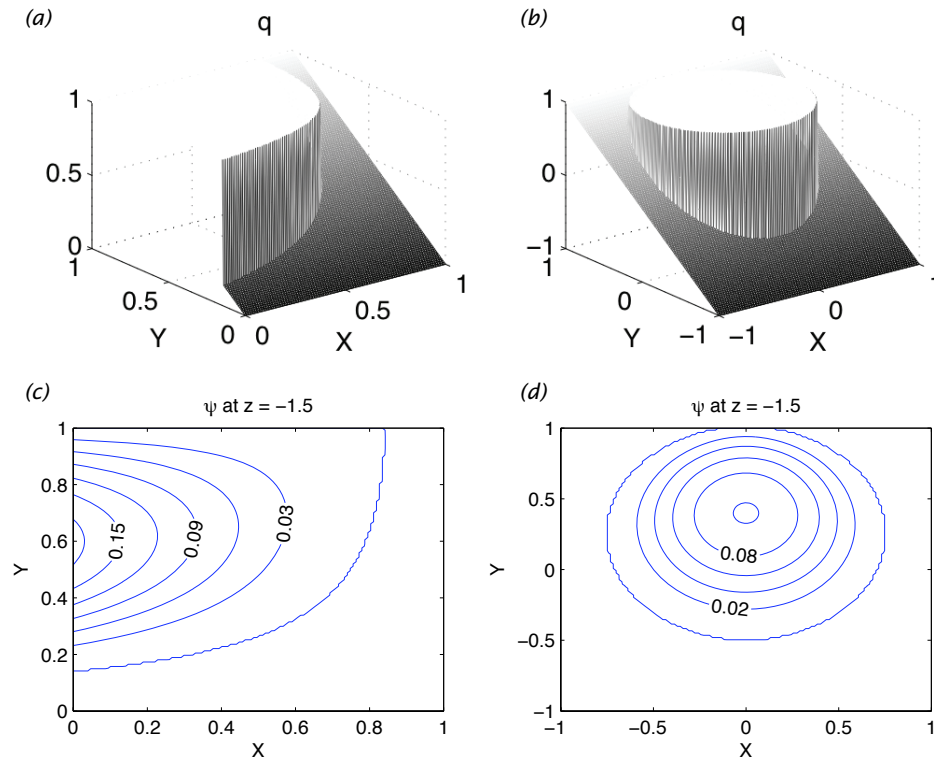


Fig. 14.18 As for Fig. 14.17, but now a perspective of potential vorticity (upper panels, obtained from (14.147)) and contours of streamfunction (lower panels, from (14.146)), at a nondimensional depth of $z = -1.5$. Within the bowls the circulation is clockwise within the bowls and the potential vorticity is uniform. Outside the bowls the fluid is stationary and the potential vorticity has the planetary value βy . The value of potential vorticity within the bowl is the planetary value at the poleward edge of the gyre, and is the same value at all depths.

Notes

- 1 This figure is based in part on earlier drawings by W. Schmitz and L. Talley that are largely based on observations, as well as on output from numerical models constrained to the observations at GFDL. However, it is highly schematic and not quantitative.
- 2 The climatological data set of Conkright et al. (2001) was assimilated into a primitive equation numerical model (MOM) by strongly relaxing the model temperature and salinity fields to those of the data set, a method arcanelly known as 'robust diagnostics' or 'nudging'. Larger scale features are broadly representative of reality but smaller scale feature may be inaccurate.
- 3 Courtesy of R. Zhang. See also Zhang and Vallis (2006).
- 4 Henry Stommel (1920–1992) was one of the most creative and influential physical oceanographers of the 20'th century. Spending most of his career at Woods Hole Institute of Oceanography, his enduring contributions include the first essentially correct theory of western intensification (and so of the Gulf Stream), some of the first models

of abyssal flow and the thermohaline circulation (chapter 15), and his foundational work on the thermocline. His *forté* was in constructing elegantly simple models of complex phenomena — often models that were physically realizable in the laboratory — and at the same time testing and encouraging the testing of the models against observations. This chapter might have been entitled ‘Variations on a theme of Stommel’.

- 5 The asymptotic solution to this boundary value problem was obtained by Wasow (1944), a few years prior to Stommel’s work, and further investigated by Levinson (1950). However, it seems unlikely these two investigators were motivated by the oceanographic problem.
- 6 Harald Sverdrup (1888–1957) was a Norwegian meteorologist/oceanographer who is most famous for the balance that now bears his name, but he also played a leadership role in scientific policy and was the director of Scripps Institution of Oceanography from 1936–1948. The Sverdrup unit is also named for him. Originally defined as a measure of volume transport, with $1 \text{ Sv} \equiv 10^6 \text{ m}^3 \text{ s}^{-1}$, it is more generally thought of as a mass transport with $1 \text{ Sv} \equiv 10^9 \text{ kg s}^{-1}$, in which case it can also be used as a measure of transport in the atmosphere. The Hadley Cell, for example, has an average transport of about 100 Sv (Fig. 11.3 and Fig. 12.21).
- 7 Leetmaa et al. (1977) and Wunsch and Roemmich (1985) offer somewhat different views on the matter.
- 8 After Munk (1950).
- 9 Courtesy of R. Zhang. See also Zhang and Vallis (2006).
- 10 Hendershott (1987), Veronis (1966a).
- 11 Veronis (1966b) investigated nonlinear effects in the Stommel and Munk problems. I am grateful to B. Fox-Kemper for providing the solutions shown in Fig. 14.9, and for many comments on this chapter. See also Fox-Kemper and Pedlosky (2003).
- 12 After Charney (1955).
- 13 A more general discussion is given by Greenspan (1962). Il’in and Kamenkovich (1964) and Ierley and Ruehr (1986) show numerically that the friction must in fact be sufficiently strong for steady boundary-layer solutions to exist. See also Fox-Kemper (2004).
- 14 After Fofonoff (1954).
- 15 Hughes and de Cuevas (2001), Jackson et al. (2006).
- 16 Welander (1968).
- 17 Figure kindly provided by Laura Jackson.
- 18 From Jackson et al. (2006).
- 19 ?.
- 20 This section and the next draw from Rhines and Young (1982b). Young and Rhines (1982) also considered the problem of a western boundary layer.
- 21 See Rhines and Young (1982a) for a numerical example of PV homogenization (numerical simulation by W. Holland). See Keffer (1985), Talley (1988) and Lozier et al. (1996) for some observed PV fields. See Holland et al. (1984) for some of both.

Further Reading

Abarbanel H. D. I. and Young, W. R., Eds., 1987. *General Circulation of the Ocean*.

This book contains several useful review articles on the oceanic general circulation. Particularly relevant to this chapter is the article by Hendershott on homogeneous single-layer models.

Pedlosky, J., 1996. *Ocean Circulation Theory*.

As well as covering the theory of wind-driven gyres, this book discusses cross-gyre flows and equatorial dynamics.

Problems

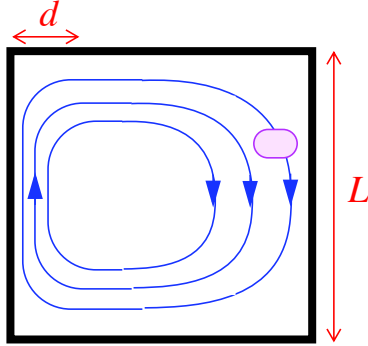
- 14.1 ♦ Obtain the exact analytic solution to the Stommel problem (14.6) in a rectangular basin, and not by way of a boundary-layer approximation. You may use a simple, single-gyre, zonal wind-stress.
- 14.2 ♦ Obtain the exact analytic solution to the Munk problem (14.42) in a rectangular basin, and not by way of a boundary-layer approximation. You may use a simple, single-gyre, zonal wind-stress.
- 14.3 Obtain a boundary-layer solution of the Munk problem, (14.44), but with ‘free-slip’ boundary conditions instead of no-slip boundary conditions. Thus, instead of setting the normal derivative of ψ to be zero at the boundaries ($\partial\psi/\partial n = 0$), set the relative vorticity at the boundary to be zero, and thus $\partial^2\psi/\partial n^2 = 0$.
- 14.4 ♦ Obtain a solution of either the Stommel or Munk problems in a triangular domain (or a circular domain).
- 14.5 What is the formal asymptotic accuracy of derivatives in the boundary-layer solution to the Stommel problem? Show that the nonlinear term $J(\psi_s, \nabla^2\psi_s)$, where ψ_s is the streamfunction, is estimated to $\mathcal{O}(1/\delta_W^3)$ accuracy, where δ_W is the width of the western boundary layer.
- 14.6 ♦ (a) Using regular perturbation theory explicitly obtain the nonlinear corrections to the Stommel solutions for a single gyre flow, and show that (14.63) is an appropriate approximation of this. Discuss the properties of the solution, and in particular, show where are the errors largest. Optional: Repeat this calculation using the exact solution to the Stommel problem, and compare the solutions to those obtained using the boundary-layer approximation. [This problem may be made easier by using algebraic manipulation software.]
 ♦ (b) Suggest a better approach to this problem that is still semi-analytic or that at least gives some insight into the nonlinear solution.
- 14.7 ♦ Do problem 14.6, but with lateral viscosity instead of bottom drag, as in the Munk problem.
- 14.8 Suppose that nonlinearity were expected to be of leading order importance in (14.58). Show that this suggests that the nondimensional version of the equation should be:

$$J(\hat{\psi}, \hat{\zeta}) + \gamma \frac{\partial \hat{\psi}}{\partial \hat{x}} = \epsilon'_S \nabla^2 \hat{\psi} + \text{curl}_z \hat{\tau}_T + \epsilon'_M \nabla^2 \hat{\zeta} \quad (\text{P14.1})$$

where $\gamma = \beta L^2/U = \beta/(|\tau|L^5)^{1/2}$ and $\epsilon'_S = r(L/|\tau|)^{1/2}$. Friction becomes less important as τ increases. Is there a limit in which friction is nowhere important?

- 14.9 Consider the two-dimensional, non-divergent flow in a box of side L , as pictured. The equatorward flow, of magnitude U , is weak and spread over a lateral scale L , whereas

the poleward flow is concentrated in a layer of width $d \ll L$. Suppose the initial condition is a spot of tracer, as shown.



Estimate how long it takes to homogenize the tracer in terms of U , L , κ and d . *Hint:* Consider the advection-diffusion equation,

$$\frac{\partial \phi}{\partial t} + \mathbf{u} \cdot \nabla \phi = \kappa \nabla^2 \phi. \quad (\text{P14.2})$$

and estimate advective and diffusive times in the western boundary layer and the interior.

- 14.10 For a two-layer wind-driven quasi-geostrophic model, obtain an expression for the minimum strength of the wind that leads to closed contours of $\bar{q} = \beta y + F\bar{\psi}$. In particular, for the solution illustrated in Fig. 14.15, at what values of A do closed contours first form?
(*Answer:* $A = \pi^{-1}$.)
- 14.11 Show by differentiating (14.146) that potential vorticity is uniform in the pool region. What is its value?